1 Lecture-1(Ring, Subring, Integral Domain, Division Ring, Field)

Definition 1.1. *A* ***ring*** *is an ordered triple* $(R, +, \cdot)$ *where* R *is a nonempty set and* $+$ *and* $\cdot$ *are binary operations on* R *satisfying the following properties:*

A1 $\forall a, b, c \in R \ \ (a + (b + c) = (a + b) + c)$.

A2 $\forall a, b \in R \ \ (a + b = b + a)$.

A3 $\exists \theta \in R(\forall a \in R \ \ (a + \theta = a))$.

A4 *For every* $a \in R$ *there is an element* $b \in R$ *such that* $a + b = 0$.

M1 $\forall a, b, c \in R \ \ (a \cdot (b \cdot c) = (a \cdot b) \cdot c)$.

D1 $\forall a, b, c \in R \ \ (a \cdot (b + c) = a \cdot b + a \cdot c)$.

D2 $\forall a, b, c \in R \ \ ((b + c) \cdot a = b \cdot a + c \cdot a)$.

Terminology

- If $(R, +, \cdot)$ is a ring, the binary operation $+$ is called *addition* and the binary operation $\cdot$ is called *multiplication*.
- *In the future, we will usually write* ab *instead of* $a \cdot b$.
- The element θ mentioned in A3 is called the **zero** of the ring. Note that we have not assumed that θ behaves like a *zero*, that is, we have not assumed that $\theta \cdot a = a \cdot \theta = \theta$ for all $a \in R$.
- What A3 says is that θ is an identity with respect to addition. Note that *negative* (as the opposite of *positive*) has no meaning for most rings. From group theory we know that θ is unique, hence we denote it as 0.
- We do not assume that multiplication is commutative and we have not assumed that there is an identity for multiplication, much less that elements have inverses with respect to multiplication.
- The element b mentioned in A4 is written $-a$ and we call it as the *additive inverse* of a (as from group theory we know that addtive inverse of an element is unique). Subtraction in a ring is defined by the rule $a - b = a + (-b)$ for all a, b in R.
- *Unless otherwise stated, from now on we will refer to the ring* R *rather than the ring* $(R, +, \cdot)$. Of course, if we define a ring, we must say what the binary operations of addition and multiplication are.

What properties that R has with respect to multiplication?

If $(R, +, .)$ is a ring, then $(R, +)$ is an abelian group. Hence with respect to addition R has the identity element, inverse of every element, commutative,cancellation laws hold etc. Thus interesting to know what extra properties that R has with respect to multiplication. That leads to various definitions.

Ring with unity

Definition 1.2. *Let R be a ring. If there is an identity with respect to multiplication, it is called the* ***identity*** *of the ring and is usually denoted by 1. If such an element exists, we say that R is a* ***ring with identity*** *or* **ring with unity**.

In some cases, the identity of a ring may be denoted by some symbol other than 1 such as e or I.

Definition 1.3. *We say that a ring R is* ***commutative*** *if the multiplication is commutative. Otherwise, the ring is said to be* ***non-commutative***.

One more time:Note that the addition in a ring is always commutative, but the multiplication may not be commutative.

Zero divisor, Unit

Definition 1.4.

- *Let R be a ring. An element a of R is called a* **zero divisor** *if there is a non-zero element b in R such that $ab = 0$ or $ba = 0$.*
- *Let R be a ring has an identity $1 \neq 0$. An element u of R is called* **unit** *in R if there is some v in R such that $uv = vu = 1$. We let $U(R)$ denote the set of all units of R. If such a v exists we write $v = u^{-1}$. We sometimes call u^{-1} the* multiplicative inverse *of u.*

Integral domain(ID)

Definition 1.5. *A ring R is said to be an* ***integral domain*** *if the following conditions hold:*

1. *R is commutative.*
2. *R contains an identity $1 \neq 0$.*
3. *If $a, b \in R$ and $ab = 0$ then either $a = 0$ or $b = 0$.*

Definition 1.6. *A ring R is said to be a* **devision ring** *(or* ***skew field****) if it satisfies the following properties.*

1. *R contains an identity $1 \neq 0$.*
2. *For each $x \in R$ such that $x \neq 0$, there is a $y \in R$ such that $xy = 1$.*

A commutative division ring is called **field**

Problem 1.7. *Which of the following are rings? If so which have identities, which are commutative, which are integral domains and which are fields? Also find $U(R)$ set of units if R is ring.*

1. $(\{0\}, +, \cdot)$.
2. $(\mathbb{N}, +, \cdot)$.
3. $(2\mathbb{Z}, +, \cdot)$ *where* $2\mathbb{Z}$ *is the set of even integers.*
4. $(\mathbb{R}, +, \cdot)$.
5. $(\mathbb{C}, +, \cdot)$.
6. $(\mathbb{Q}, +, \cdot)$.
7. $(\mathbb{Z}, +, \cdot)$.
8. $(\mathbb{Z}_2, +, \cdot)$.
9. $(\mathbb{Z}_3, +, \cdot)$.
10. $(\mathbb{Z}_4, +, \cdot)$.
11. $(M_2(\mathbb{R}), +, \cdot)$.
12. $(M_2(\mathbb{Z}_n), +, \cdot)$.
13. $(M_n(\mathbb{R}), +, \cdot)$.

Definition 1.8. *If R is a ring, $a \in R$ and $n \in \mathbb{N}$ we define a^n by the following rules:*

$a^1 = a$,

$a^n = aa \cdots a$ *(n copies of a) if $n \geq 2$.*

If R has an identity 1 and a is a unit then we can also define:

$a^0 = 1$,

a^{-1} = *multiplicative inverse of a,*

$a^{-n} = (a^{-1})^n$ *for $n \geq 2$.*

Note that since generally an element a of a ring is not a unit, we cannot expect a^n to be defined for negative integers.

Problem 1.9. *Which of the following are rings? If so which have identities, which are commutative, which are integral domains and which are fields? Also find $U(R)$ set of units if R is ring.*

1. *Let R be any commutative group (denoting the group operation by $+$) and defining the multiplication $\cdot$ on R as $a \cdot b = 0$ for all $a, b \in R$.*

2. *(The(real) Hamiltonian Quaternions) Let $H = \{a + bi + bj + dk | a, b, c, d \in \mathbb{R}\}$ where*
   a) addition is defined by component wise i.e., $(a_1 + b_1 i + c_1 j + d_1 k) + (a_2 + b_2 i + c_2 j + d_2 k) = (a_1 + a_2) + (b_1 + b_2)i + (c_1 + c_2)j + (d_1 + d_2)k$
   b)Multiplication is defined by expanding $(a_1 + b_1 i + c_1 j + d_1 k) \cdot (a_2 + b_2 i + c_2 j + d_2 k)$ using distributive law and the following relations
   $\boxed{i^2 = j^2 = k^2 = -1,\ ij = k, jk = i, ki = j \text{ and } ji = -k, kj = -i, ik = -j}$

3. *Let X be any non-empty set and A be any ring. Let*
$$R_X^A = \{f | f : X \to A\}.$$
   That is R_X^A be the collection of all functions from X to A, where addition and multiplication defined component wise i.e.,
$$(f + g)(x) = f(x) + g(x) \text{ and } (fg)(x) = f(x)g(x).$$

4. *Let $R_{[0,1]}^{\mathbb{R},cont}$ be set of all continuous functions from $[0, 1]$ to $\mathbb{R}$, where addition and multiplication defined component wise.*

5. *Let $R_{[0,1]}^{\mathbb{R},diff}$ be set of all differentiable functions from $[0, 1]$ to $\mathbb{R}$, where addition and multiplication defined component wise.*

6. *Let $R_{\mathbb{R}}^{\mathbb{R},CS}$ be set of all functions from $\mathbb{R}$ to $\mathbb{R}$ with compact support, where addition and multiplication defined component wise.*
   A function $f : \mathbb{R} \to \mathbb{R}$ is said to have compact support if there are real numbers a and b (depending on f) such that $f(x) = 0$ for all $x \notin [a, b]$ (i.e., f is zero outside some bounded interval).

7. *Let D be a rational number that is not a perfect square in $\mathbb{Q}$ and define*

$$\mathbb{Q}(\sqrt{D}) = \{a + b\sqrt{D} | a, b \in \mathbb{Q}\}.$$

   Addition and multiplication defined as

   - $(a + b\sqrt{D}) + (c + d\sqrt{D}) = (a + c) + (b + d)\sqrt{D}$.
   - $(a + b\sqrt{D}) \cdot (c + d\sqrt{D}) = (ac + Dbd) + (ad + bc)\sqrt{D}$.

8. *(Gaussian Integers) $\mathbb{Z}[i] = \{a + ib | a, b \in \mathbb{Z}\}$. This is similar to complex numbers $\mathbb{C}$ but coefficients are taken from integers rather real numbers.*

9. *Let R be a commutative ring with unity $1 \neq 0$ and let $G = \{g_1, g_2, \ldots, g_n\}$ be any finite group operation written multiplicatively. Define group ring RG as*

$$RG = \left\{ \sum_{i=1}^{n} a_i g_i | a_i \in R \right\}$$

   addition and multiplication defined as

$$\left(\sum_{i=1}^{n} a_i g_i \right) + \left(\sum_{i=1}^{n} b_i g_i \right) = \sum_{i=1}^{n} (a_i + b_i) g_i$$

   and

$$\left(\sum_{i=1}^{n} a_i g_i \right) \cdot \left(\sum_{i=1}^{n} b_i g_i \right) = \sum_{k=1}^{n} \sum_{g_i g_j = g_k} (a_i b_j) g_k.$$

10. *Let R be a commutative ring with unity $1 \neq 0$. Let*

$$R[x] = \{a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n | a_0, \ldots, a_n \in R, \text{ with } n \in \mathbb{Z}^+ \cup \{0\}\}$$

   be the set of all polynomials with coefficients from R with usual polynomial addition and multiplication.

11. *Let X be a non-empty set. Let $P(X)$ denote the power set (set of all subsets of X), where addition $(+)$ is the symmetric difference of sets and multiplication $(\cdot)$ is intersection of sets.*

12. *Let R be a commutative ring with unity. Define the set $R[[x]]$ of formal power series in the indeterminate x with coefficients from R to be all formal infinite sums.*

$$R[[x]] = \left\{ \sum_{i=1}^{\infty} a_i x^i | a_i \in R \right\}$$

   where addition and multiplication is defined as

$$\left(\sum_{i=1}^{\infty} a_i x^i \right) + \left(\sum_{i=1}^{\infty} b_i x^i \right) = \sum_{i=1}^{\infty} (a_i + b_i) x^i$$

and

$$\left(\sum_{i=1}^{\infty} a_i x^i\right) \cdot \left(\sum_{i=1}^{\infty} b_i x^i\right) = \sum_{i=1}^{\infty} \left(\sum_{k=0}^{n} (a_k b_{n-k}) x^k\right).$$

Definition 1.10. *A subset S of a ring R is said to be a* ***subring*** *of R if the following conditions hold:*

1. $0 \in S$.

2. *If $a \in S$, then $-a \in S$.*

3. *If $a, b \in S$, then $a + b \in S$ and $ab \in S$.*

If R is a field and the following conditions also hold:

4. $1 \in S$.

5. *If $a \neq 0$ and $a \in S$, then $a^{-1} \in S$.*

we say that S is a ***subfield*** *of R.*

If S is a subring (subfield) of the ring (field) R, then it is easy to verify that S is itself a ring (field) with respect to the addition and multiplication on R.

Problem 1.11. *1. Let R be a ring. Then*

(a) $0a = a0 = 0$ *for all* $a \in R$.

(b) $(-a)(b) = (a)(-b) = -(ab)$ *for all* $a, b \in R$.

(c) $(-a)(-b) = ab$.

(d) $a(b - c) = ab - ac$.

(e) $(b - c)a = ba - ca$.

(f) If R has identity element 1, then i)the identity is unique ii) $(-a) = (-1)a$ *iii)* $(-1)^2 = 1$ *iv) If u is unit in R then so is $-u$.*

2. (Subring Test) A non-empty set S of ring R is subring if S is closed under subtraction and multiplication.

3. Let R be a ring with unity. Find two non-trivial subrings of $M_n(R)$.

4. Let R and S be rings. Prove that the direct product $R \times S$ is a ring under component wise addition and multiplication.

5. Prove that cancellation laws hold in integral domain.

Note that cancellation laws always hold with respect to addition. Let R be an integeral domain, suppose $a \neq 0$ and $ab = ac$. Then we have $a(b-c) = 0$ (we missed few steps). Since R is ID, $b - c = 0$. Is the converse true? That is if R is commutative ring with unity. Also suppose cancellation laws are hold in R, then is R an ID?
Let us suppose $a \neq 0$ and $ab = 0$. Then $ab = a.0$. By LCL, $b = 0$.
Conclusion: Let R be a commutative ring with unity. Then R is ID if and only if cancellation laws are hold in R.

6. *Every field is an integral domain.*

   Let R be a field. Then by definition R is a commutative ring with unity. Let $ab = 0$. If $a = 0$ then we are done. Suppose $a \neq 0$, then a^{-1} exsits in R and $a^{-1}(ab) = a^{-1}0$. Hence $b = 0$.

7. *Every finite integral domain is a field.*

   Let R be a finite integral domain. Then R is a a commutative ring with unity. We need to show that every nonzero element has an inverse. Let $a \in R \setminus \{0\}$. Consider the function $f_a : R \to R$ defined as $f_a(x) = ax$. Show that f is one-one. Since R is finite f_a is onto hence 1 has a preimage.

8. *A ring R is called a Boolean ring if $a^2 = a$ for all $a \in R$. Prove that every Boolean ring is commutative.*

   Since $(a+a)^2 = a+a$, we have $a = -a$. Then consider $(a+b)^2 = a+b$. Hence $ab = -(ba) = ba$. Note that $2a = a + a = a + (-a) = 0$.

9. *Prove that only Boolean ring that is integral domain is $\mathbb{Z}_2$.*

10. *Prove that $\mathbb{Z}_n$ is a field if and only if it is an integral domain.*

11. *Prove that $\mathbb{Z}_n$ is a field if and only if n is a prime.*

12. *Let R be a ring with identity. The center of a ring R is $\{z \in R | zr = rz \forall r \in R\}$ (i.e., set of all elements that commute with every element of R). Prove that center of ring is subring that contains the identity. Prove that center of division ring is field.*

13. *Let R be a ring with identity. For a fixed element $a \in R$ define*

    $$C(a) = \{r \in R | ra = ar\}.$$

    Prove that $C(a)$ is subring of R containing a. Prove that center of ring is the intersection of $C(a)$ for all elements over $a \in R$.

14. *If D is a division ring, then $C(a)$ is a division ring for $a \in D$.*

15. *R be an integral domain and $p(x), q(x) \in R[x]$. Then*

(a) $deg(p(x)q(x)) = deg(p(x)) + deg(q(x))$

(b) *The units of* $R[x]$ *are simply the units of* R.

(c) $R[x]$ *is also an integral domain.*

16. *Let* R *be a commutative ring with unity* $1 \neq 0$ *and let* $G = \{g_1, g_2, \ldots, g_n\}$ *be any finite group operation written multiplicatively. Then show that*

$$N = g_1 + g_2 + \cdots + g_n$$

is in the center of the group ring RG.

17. *Let* S *be a subring of ring* R. *Then provide an example for the following cases.*

(a) S *may be commutative but not* R.

(b) S *and* R *may both have unity but they may not be the same.*

(c) S *may not have unity even if* R *has.*

(d) S *may have unity but* R *may not.*

(e) *Assume* R *and* S *have same unity. And an element* $a \in S$ *is unit in* R *but not in* S.

(f) *If the unities in* R *and* S *exists and are different, then an element* $a \in S$ *may be a unit in* S *but not in* R.

18. *Show that the set* $\left\{ \begin{bmatrix} a & b \\ \overline{b} & a \end{bmatrix} | a, b \in \mathbb{C} \right\}$ *is not a division ring where* $\mathbb{C}$ *is the field of complex numbers and* $\overline{z}$ *denote complex conjugate of* z. *Find all units in this ring.*

19. *If* R *is commutative ring with* 1, *then* $A \in M_n(R)$ *is a unit if and only if its determinant,* $det(A)$ *is a unit in* R.

20. *Give an example of Boolean ring without unity.*

21. *A ring in which the elements* 1 *and* 0 *are equal is the zero ring.*

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2 Lecture-2 (Ideals,Quotient Rings and Homomorphism of Rings)

Definition 2.1. *(Characteristic of ring) The* **characteristic** *of ring R denoted $char(R)$ is the least positive integer n such that $nx = 0$ for all $x \in R$. If no such integer exists, then we say R has characteristic zero.*

Problem 2.2.

1. *Show that ring of integers has characteristic zero. If a ring R has characteristic zero, then any ring which contains R also has characteristic zero.*
2. *The characteristic of integral domain is either zero or prime.*
3. *Show that a ring with finite number of elements has finite characteristic.*
4. *Give an example of an integral domain with infinite number of elements but of finite characteristic.*
5. *Show that Boolean ring has characteristic two.*
6. *Let R be ring with characteristic two. Let $a, b \in R$ and $ab = ba$. Then $(a+b)^2 = (a-b)^2$.*
7. *Let R be a ring with unity and k is the least positive integer such that $k \cdot 1 = 0$. Then char(R)=k.*

Definition 2.3. *Let R be a ring, let I be a subset of R and let $r \in R$.*

1. *$rI = \{ra | a \in I\}$ and $Ir = \{ar | a \in I\}$.*
2. *A subset I of R is said to be* **left ideal** *of R if*
   (a) *I is a subring of R and*
   (b) *I is closed under left multiplication that is $Ir \subseteq I$ for all $r \in R$*

   Similarly we can define **right ideal**.
3. *A subset I that is both left ideal as well as right ideal is called* **two sided ideal** *or simply* **ideal** *of R.*

Definition 2.4. *An ideal M is an arbitrary ring R is said to be* **maximal** *ideal if $M \neq R$ and the only ideals containing M are M and R.*

- In other words, an ideal of R is a maximal ideal if is impossible to squeeze an ideal between it and the full ring.
- A maximal ideal I in the ring R is a proper ideal ($I \neq R$) (that is not contained in any strictly larger proper ideal.
- One can prove that every proper ideal in a ring with unity is contained in a maximal ideal consequently, every ring with unity has at least one maximal ideal, this can be proved, assuming a basic axiom of mathematics, the so called **axiom of choice.**

- There may be many distinct maximal ideals in a ring.

Definition 2.5. *Let R be a commutative ring. An ideal P is called a* **prime** *ideal if $P \neq R$ and whenever the product ab of two elements $a, b \in R$ is in P then at least one of a and b is an element of P.*

Definition 2.6. *Let I and J be ideals of ring R.*

1. *Denote the sum of I and J be $I + J$ is defined as*

$$I + J = \{a + b | a \in I, b \in J\}.$$

2. *Denote the product of I and J be IJ is defined to be set of all finite sums of elements of the form ab with $a \in I, b \in J$. That is*

$$IJ = \{\sum_{i=1}^{k} a_i b_i | a_i \in I, b_i \in J, \text{ for some } k, 1 \leq i \leq k\}.$$

3. *For any $n \geq 1$, define nth power of I denoted by I^n, to be the set consisting of all finite sums of elements of the form $a_1 a_2 \cdots a_n$, where $a_j \in I$ for all j.*

4. *$I \cap J = \{a : a \in I, a \in J\}$ is an ideal of $\mathcal{R}$, called the intersection of I and J.*

5. *I and J are said to be maximal if $I + J = R$.*

6. *$I : J$ is called colon ideal which is analog of division and is defined as $I : J = \{x \in R | xJ \subseteq I\}$.*

Since additive group of a ring R is commutative, every subgroup of R is a normal subgroup. Thus any ideal J of R is an invariant sub group under addition. Consequently, $R/J = \{r + J | r \in R\}$ (set of all distinct cosets of J in R.) is the quotient group with addition defined by $(r + J) + (s + J) = (r + s) + J$. Now we can define multiplication by

$$(r + J) \cdot (s + J) = (r \cdot s) + J.$$

One can easily verify that $(R/J, +, \cdot)$ is a ring called **quotient (or factor)** ring.

Problem 2.7. 1. *Let I be an ideal of a ring R, then $a + I = b + I$ if and only if $a - b \in I$, where $a, b \in R$.*

Proof. Let $a + I = b + I$. Since $a \in a + I = b + I$, hence $a = b + k$ for some $k \in I$. Conversely, suppose that $a - b \in I$ claim $a + I = b + I$.
Let $a + x \in a + I$, then $a - b + x \in I$. $a - b + x = k$ for some $k \in I$ or $a + x = b + k \in b + I$. Equivalently, $a + I \subseteq b + I$. Similarly we can prove $b + I \subseteq a + I$. $\square$

2. *If U is an ideal of ring R with unity and $1 \in U$, then show that $U = R$.*

3. *If $\mathbb{F}$ is a field, prove that its only ideals are (0) and F itself.*

4. *Let R be a commutative ring with unity and its only ideals are (0) and R. Then R is a field.*

5. *Let R be a commutative ring and $a \in R$,*

   (a) *show that $aR = \{ar | r \in R\}$ is a two sided ideal of R.*

   (b) *Show by an example that this may be false if R is not commutative.*

6. *Let $R_{[0,1]}^{\mathbb{R},cont}$ be ring of all continuous functions from $[0,1]$ to $\mathbb{R}$, where addition and multiplication defined component wise. Let $\alpha \in [0,1]$. Then*

$$M_\alpha = \{f \in R_{[0,1]}^{\mathbb{R},cont} | f(\alpha) = 0\}$$

   is a maximal ideal of $R_{[0,1]}^{\mathbb{R},cont}$. Also show that every maximal ideal of $R_{[0,1]}^{\mathbb{R},cont}$ is of this form.

7. *Find all maximal ideals in the ring of integers.*

8. *If R is a commutative ring with unity and M be an ideal of R. Then M is maximal if and only if R/M is a field.*

9. *If R is a commutative ring and M be an ideal of R. Then M is prime ideal if and only if R/M is an integral domain.*

10. *Prove that every maximal ideal in commutative ring with unity is prime but not the converse.*

11. *Let R be a finite commutative ring with unit element. Then prove that every prime ideal of R is a maximal ideal.*
    Solution:Let I be a prime ideal of R. Then R/I be an ID and finite hance R/I is a field.

12. *Let R be a commutative ring with unity. Then $\{0\}$ is prime ideal if and only if R is an integral domain.*
    Comment: Characterizing given a commutative ring with unity an ID or not by using zero ideal.

13. *Let R be a commutative ring with unity. Let P be a prime ideal. Then P can't be intersection of two ideals which properly contains P.*

    Proof. Let I, J are two ideals of R such that $P \subset I, J$ and $P = I \cap J$. Since P is a proper subset of I there exists $a \in I$ but $a \notin P$ similarly $b \in J$ but $b \notin P$. Which is a contradiction for $ab \in I \cap J = P$ and P is prime ideal. □

14. *If R is a commutative ring, let $N = \{x \in R | x^n = 0$ for some integer $n\}$. Prove*

    (a) *N is an ideal of R.*

    (b) *In R/N if $\overline{x}^m = 0$ for some m, then $\overline{x} = 0$.*

15. *Let $R = \{(a_n) : a_n \in \mathbb{R},\ (a_n)$ is a convergent sequence$\}$ and $I = \{(a_n) \in R | a_n = 0$ for all most all $n\}$. Then show that R is a ring and I is an ideal of R.*

16. *Let R be a commutative ring and suppose that A is an ideal of R. Let $r(A) = \{x \in R | x^n \in A$ for some $n\}$. Prove*

(a) $r(A)$ *is an ideal of* R *which contains* A.

(b) $r(r(A)) = r(A)$.

$r(A)$ *is often called the* **radical** *of* A.

17. *Prove or disprove center of ring* R *is an ideal of* R.

18. *Prove or disprove the set of all differentiable functions in the ring* $R^{\mathbb{R}}_{\mathbb{R}}$ *is an ideal.*

19. *If* I *and* J *are ideals of a ring* R*, then show that* $\{ab | a \in I, b \in J\}$ *need not be an ideal. Show that* $IJ \subseteq I \cap J$.

20. *Let* R *be a ring and* $Z(R)$ *denoted set of zero divisors of* R*. Find* $Z(\mathbb{Z}_n)$.

21. *Let* $R = M_n(\mathbb{R})$*. If* $A \in R$ *is not a unit, then show that* A *is a zero divisor.*

22. *If* R *is an integral domain, then* $R \times R$ *is not an integral domain, where addition and multiplication defined in component wise.*

23. *Let* $R = \{(a_n) : a_n \in \mathbb{R},\ (a_n)$ *is a Cauchy sequence*$\}$ *and* $I = \{(a_n) \in R | \lim_{n\to\infty} a_n = 0\}$*. Then show that* R *is ring and* I *is an ideal in* R.

Definition 2.8. *Let* $(R, +, \cdot)$ *and* $(S, \oplus, \odot)$ *be two rings. A function*

$$f : R \to S$$

is a ***homomorphism*** *if for all* $a, b \in R$ *we have*

$$\begin{aligned} f(a \cdot b) &= f(a) \odot f(b) \\ f(a + b) &= f(a) \oplus f(b). \end{aligned}$$

If also f *is one-to-one and onto we call* f *an* ***isomorphism****. In this case we say* R *and* S *are* ***isomorphic*** *and write* $R \cong S$.

Although it will usually be clear from the context, now that we have homomorphisms for both groups and rings, sometimes we will say *ring homomorphism* or *group homomorphism* to be specific. Similarly, for isomorphisms.

As in the case of groups, if two rings are isomorphic, then they share almost all properties of interest. For example, if R and S are isomorphic rings, then R is a field if and only if S is a field.

It can be easily verified that if $\mathbf{0}_1$ is the zero element of $\mathcal{R}_1$ and $\mathbf{0}_2$ is the zero element of $\mathcal{R}_2$ then for every homomorphism $f : \mathcal{R}_1 \to \mathcal{R}_2$, $f(\mathbf{0}_1) = \mathbf{0}_2$. Thus, the set $\ker(f) = \{a \in \mathcal{R}_1 : f(a) = \mathbf{0}_2\}$ is non-empty and is called the *kernel* of f.

Problem 2.9. 1. *Let* $\mathcal{R}_1$ *and* $\mathcal{R}_2$ *be two rings and let* $f : \mathcal{R}_1 \to \mathcal{R}_2$ *be a ring homomorphism. Then*

(a) $f(\mathcal{R}_1) = \{f(a) : a \in \mathcal{R}_1\}$ *is a subring of* $\mathcal{R}_2$,

(b) *ker*(f) *is an ideal of* $\mathcal{R}_1$*, and*

(c) *(First Isomorphism Theorem)* $\mathcal{R}_1/ker(f) \cong f(\mathcal{R}_1)$.

2. *For $n \in \mathbb{Z}$, the map $f_n : \mathbb{Z} \to \mathbb{Z}$ defined by $f_n(x) = nx$ is homomorphism only if $n = 0$ or $n = 1$. But f_n is always a group homomorphism.*

3. *For each fixed $c \in X$, the map $E_c : R_X^A \to A$ defined by $E_c(f) = f(c)$ (called* **evaluation** *at c) is a ring homomorphism. Find ker E_c and $R_X^A/ker\ E_c$.*

4. *Let R be a ring. Then the map from polynomial ring $R[x] \to R$ defined by $p(x) = p(0)$ (evaluation at 0) is a ring homomorphism.*

5. *Let R be a ring and J is an ideal of R. Then show that the mapping $f : R \to R/J$ defined by $f(a) = a + J$ is a homomorphism and onto.*

6. *Show that every field contains either a subfield isomorphic to $\mathbb{Q}$ or to $\mathbb{Z}_p$, where p is prime. [The fields $\mathbb{Q}$ and $\mathbb{Z}_p$ are called* **prime** *fields.]*

7. *Let R be an integral domain of characteristic p, where p is a prime. Prove the map $R \to R$ defined by $x \to x^p$ is a ring homomorphism (It is called* **Frobenius** *homomorphism).*

8. *Given any integral domain D, there is a filed $\mathbb{F}$ containing an isomorphic image of D as subring.*

9. *Let X be a non-empty set and let $P(X)$ be the power set of X. Let R be set of all functions from X into $\mathbb{Z}_2$. For each $A \in P(X)$ defines the function*

$$\chi_A : X \to \mathbb{Z}_2 \text{ by } \chi_A(x) = \begin{matrix} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{matrix}.$$

(χ_A is called **characteristic** *function of A with values in $\mathbb{Z}/2\mathbb{Z}$). Prove that the map $P(X) \to R$ defined $A \to \chi_A$ is a ring homomorphism.*

10. *Show that the identity map is the only ring automorphism in the filed of real numbers.*

11. *Let $\mathbb{F}$ be a filed. If the characteristic of $\mathbb{F}$ is zero, show that $\mathbb{F}$ has a subfield isomorphic to $\mathbb{Q}$. If the characteristic of $\mathbb{F}$ is p for some prime number p, show that $\mathbb{F}$ has a subfield isomorphic to $\mathbb{Z}_p$.(hint: define a map $f : \mathbb{Z} \to \mathbb{F}$ by $f(n) = n \cdot 1_{\mathbb{F}}$ and find kernel of f.)*

12. *Let R be a ring and $f : R \to R$ a ring homomorphism. For parts (c) and (d), assume there exists a positive integer N such that $f^{N+1}(R) = f^N(R)$ and $ker(f^N) = ker(f^{N+1})$.*
    (a) Prove that $f^{N+1}(R) \subseteq f^N(R)$ for all positive integers n.
    (b) Prove that $ker(f^N) \subseteq ker(f^{N+1})$ for all positive integers n.
    (c) Prove $f^N(R) \cup ker(f^N) = \{0\}$.
    (d) Prove $f^N(R) + ker(f^N) = R$.

13. *(The Second Isomorphism Theorem of Rings) Let A be a subring and let B be an ideal of R. Then $A + B = \{a + b | a \in A, b \in B\}$ is subring of R, $A \cap B$ is an ideal of A and*

$$A + B/B \cong A/A \cap B.$$

14. *(The Third Isomorphism Theorem of Rings) Let R be a ring and $I \subseteq J$ ideals of R. Show that J/I is an ideal of R/I and $R/J \cong (R/I)/(J/I)$ as rings.*

15. *(The Fourth Isomorphism Theorem of Rings) Let R be a ring. Let I be an ideal of of R. The correspondence $A \leftrightarrow A/I$ is an inclusion preserving bijection between the set of subrings A of R that contain I and the set of subrings of R/I. Furthermore, A (a subring containing I) is an ideal of R if and only if A/I is an ideal of R/I.*

16. *Show that identity function is the only automorphism in $\mathbb{Z}$.*

    Proof. Let $a = f(1)$. Then $a^2 = f(1)^2 = f(1)f(1) = f(1.1) = f(1) = a$ so $a(a-1) = 0$. Also f is injective so $a \neq 0$. Hence we have $f(1) = 1$
    Now let $b \in \mathbb{N} \Rightarrow f(b) = f(1+1+1+1+\cdots+1) = f(1)+f(1)+\cdots+f(1) = bf(1) = b$. $f(-b) = -f(b) = -b$. □

17. *If R is a commutative ring with unity, then show that there exists a maximal ideal for R.*

18. *Find few rings which are properly contained in $\mathbb{Q}$ but properly contains $\mathbb{Z}$.*

19. *Find few rings which are properly contained in $\mathbb{Q}[x]$ but properly contains $\mathbb{Z}[x]$.*

20. *Let $m, n \in \mathbb{Z}$. Find $m\mathbb{Z} : n\mathbb{Z}$.*

21. *Let R be ring and I is an ideal of R. Then find $I : R$ and $R : I$.*

22. *Let R be a commutative ring with unity and $S \subseteq R$. Then*

    $$Ann_R(S) = \{r \in R | \forall s \in S(rs = 0)\}.$$

    Show that $Ann_R(S)$ is an ideal of R. If $S = \{x\}$, then we denote $Ann_R(S)$ as $Ann_R(x)$ or simply $Ann(x)$ if R is clear from the context.

23. *Let R be a Boolean ring with unity. Then show that only nilpotent elements in R is the zero element.*

24. *Let R be a ring with 1. Suppose that a, b are elements in R such that $ab = 1$ and $ba \neq 1$.*

    (a) *Prove that $1 - ba$ is idempotent.*

    (b) *Prove that $b^n(1 - ba)$ is nilpotent for each positive integer n.*

    (c) *Prove that the ring R. has infinitely many nilpotent elements.*

    Solution:First note that $a^n b^n = aa \cdots abbb \cdots b = 1$. If $m > n$, then $a^m b^n = b^{m-n}$. To prove part c show that $b^n(1 - ba) \neq b^m(1 - ba)$ for each pair of distinct integers m, n by using part b.

3 Lecture-3 (Field of Fractions, PIDs and UFDs)

- Let R be a commutative ring. If an element $a \in R$ is neither zero nor zero divisor and $ab = ac$ then $b = c$.
- Thus a non-zero element that is not a zero divisor enjoys some of the properties of unit without necessary possessing a multiplicative inverse in R.
- Now we construct a ring which contains R and all non-zero non-zero divisor elements in R become units in that ring. In particular, if R is an integral domain then that extended ring is field called **Field of Fractions.**

Theorem 3.1. *Let R be a commutative ring. Let D by any non-empty subset of R that does not contain 0, does not contain any zero divisors and is closed under multiplication. Then there is a commutative ring Q with 1 such that Q contains R as a subset and every element of D is a unit in Q. The ring Q has the following additional properties.*

1. *Every element in Q is of the form rd^{-1} for some $r \in R$ and $d \in D$. In particular Q is field if $D = R \setminus \{0\}$.*
2. *(Uniqueness of Q) The ring Q is the "smallest" ring containing R in which all elements of D become units, in the following sense. Any ring containing an isomorphic copy of R in which all elements of D become units must also contain an isomorphic copy of Q.*

Definition 3.2. *Let R be a ring with identity 1($1 \neq 0$).*

1. *Let $\langle A \rangle$ denote the smallest ideal of R containing A, called the* **ideal generated by** *A.*

$$\langle A \rangle = \cap_{I \text{ an ideal } A \subseteq I} I.$$

2. *An ideal generated by a single element x denoted by $\langle x \rangle$ is called* **principal ideal generated by** *x.*
3. *An ideal generated by a finite set is called a* **finitely generated** *ideal.*
4. *A* **principal ideal domain (P.I.D)** *is an integral domain in which every ideal is principal.*

Problem 3.3. 1. *Show that $\mathbb{Z}$ is a PID.*

Proof. Let I be an ideal of $\mathbb{Z}$. If $I = \{0\}$, then nothing to show. Suppose that $I \neq \{0\}$, consequently $0 \neq a \in I$. Or equivalently $\langle a \rangle \subseteq I$. Thus $\mathcal{S} = I \cap \mathbb{Z}^+$ is a non-empty set as $|a| \in \mathcal{S}$. Hence by well ordering principle $\mathcal{S}$ has a least element say d.
Now we claim $I = \langle d \rangle$. It is clear that $\langle d \rangle \subseteq I$. Suppose $y \in I$. Since $d \neq 0$ hence by devision algorithm, there exists unique $q, r \in \mathbb{Z}$ such that $y = dq + r$, where $0 \leq r < d$. Thus $r = 0$ follows from fact that $r = y - dq \in I$ and d is the least positive in I. Thus $y \in \langle d \rangle$ or $I \subseteq \langle d \rangle$. $\square$

2. *Show that every field is a PID.*

Proof. As we know every field $\mathbb{F}$ contains only two Ideals $I = \{0\} = \langle 0 \rangle$ and $\mathbb{F} = \langle 1 \rangle$ and both are principal. $\square$

3. *Show that $\mathbb{Z}[x]$ is not a PID.*

Proof. First note that $\langle 2, x \rangle = \{2f(x) + xg(x) | f(x), g(x) \in \mathbb{Z}[x]\}$ be the set of polynomials generated by 2 and x. Also one can see that this is exactly the set of all polynomials with even constant coefficients. One can easily check that it is a proper ideal of $\mathbb{Z}[x]$.
We claim that this ideal is not a principal ideal. On contrary assume $\langle 2, x \rangle = \langle h(x) \rangle$, for some $h(x) \in \mathbb{Z}[x]$.
Consequently $2 \in \langle h(x) \rangle$ or $2 = h(x)f(x)$ for some $f(x) \in \mathbb{Z}[x]$.

$$0 = \deg(2) = \deg(h(x)) + \deg(f(x))$$

hence both $f(x), h(x) \in \mathbb{Z}$ and 2 is prime in $\mathbb{Z}$. Hence $h(x), f(x) \in \{\pm 2, \pm 1\}$. Now if $h(x) = \pm 1$, then $\langle 2, x \rangle = \mathbb{Z}[x]$ which is a contradiction for $\langle 2, x \rangle$ is a proper ideal of $\mathbb{Z}[x]$. But $h(x) = \pm 2$ is not possible as $x \in \langle 2, x \rangle$ and $x \notin \langle h(x) \rangle$. $\square$

4. *Show that $R^{\mathbb{R}}_{[0,1]}$ is not PID.*

Proof. We know that it is not integral domain. $\square$

5. *Let R be a commutative ring with unity. Prove that an ideal generated by x in the polynomial ring $R[x]$ is prime ideal if and only if R is an integral domain. Prove that (x) is a maximal ideal if and only if R is a field.*

Definition 3.4. *Let R be an integral domain.*

1. *A non-zero, non-unit element $a \in R$ is said to* **divide** *another element $b \in R$ if $b = aq$ for some non-unit $q \in R$.*

2. *If $a, b \in R$, then $d \in R$ is said to be a* **greatest common divisor** *of a and b if*

   (a) *$d|a, d|b$,*

   (b) *whenever $e|a, e|b$, then $e|d$.*

3. *Suppose $r \in R$ is non-zero and is not a unit. Then r is called* **irreducible** *in R if whenever $r = ab$ with $a, b \in R$, at least one of a or b must be a unit in R otherwise r is said to be* **reducible**.

4. *The non-zero $p \in R$ is called* **prime** *in R if the ideal $\langle p \rangle$ generated by p is prime ideal. In other words, a non-zero element p is prime if it is not a unit and whenever $p|ab$ for any $a, b \in R$, then either $p|a$ or $p|b$.*

5. *Two elements a and b are called* **associates** *if each divides the other, in other words, there exists a unit u such that $a = ub$.*

Definition 3.5. *An integral domain, R with unit element is a* **unique factorization domain** *if*

1. *Any nonzero element in R is either a unit or can be written as product of finite number of irreducible elements in R.*

2. *The decomposition in part (1) is unique up to the order and associates of the irreducible elements.*

Problem 3.6. 1. *If R is a unique factorization domain, then so is $R[x]$. Use this to show that every UFD need not be a PID.*

2. *Let R be an integral domain. Prove that $\langle a \rangle = \langle b \rangle$ for some elements $a, b \in R$, if and only if $a = ub$ for some unit u of R (in other words a and b are associates).*

Proof. First suppose $a = ub$ that is a and b are associates. So $b = u^{-1}a$, $a \in \langle b \rangle$ and $b \in \langle a \rangle$. In other words $\langle a \rangle = \langle b \rangle$.
Conversely, suppose that $\langle a \rangle = \langle b \rangle$. So we have $a \in \langle b \rangle$ or $a = bx$ for some $x \in R$ similarly $b = ay$ for some $y \in R$. Or equivalently $a = bx = ayx \Rightarrow xy = 1$. □

3. *Let R be a ring with unit. Define a relation $\sim$ on R as $a \sim b \Leftrightarrow a$ and b are associates. Then show that $\sim$ is an equivalence relation on R.*

4. *Let R be a commutative ring with identity. An element $a \in R$ is said to be* **nilpotent** *if there exists a natural number n such that $a^n = 0$.*

   (a) *Show that if $x \in R$ is nilpotent, then $1 + xy$ is a unit of R for every $y \in R$. (hint: expand the expression $(1 - (-xy)^n)$.)*

   (b) *Show that the set N of all nilpotent elements of R is an ideal of R.*

   (c) *Show that the quotient ring R/N contains no non-zero nilpotent elements.*

5. *Show that in an integral domain, every prime element is irreducible.*

Proof. Let $a \in R$ be a prime element. By definition of prime element a is non-zero and non-unit. Claim: a is irreducible. Suppose $a = bc$. Then $a|bc$ so $a|b$ or $a|c$ as a is prime. If $a|b$, then $b = ax$ for some $x \in R$. Thus $a = (ax)c \Rightarrow xc = 1$. That is c is a unit. Similarly if $a|c$, then we can show that b is unit. □

6. *In a UFD a non-zero element is a prime if and only if it is irreducible.*

Proof. We already know that in integral domain every prime is irreducible. Suppose a is an irreducible element in a UFD R. Suppose $a|bc$, then $ax = bc$ for some $x \in R$. Since R is unique factorization domain, b, c are expressed as product of irreducible elements uniquely. So a must be associate of one of the irreducible element. Hence a divides b or c.
($ax = p_1p_2 \cdots p_k \cdot q_1q_2 \cdots q_l$, [Further R is UFD] so but a is irreducible consequently from definition of irreducibility a is one of p_i or one of q_j.) □

7. *In a PID a non-zero element a is irreducible if and only if $\langle a \rangle$ is a maximal ideal.*

Proof. Let a be an irreducible element in R. Then a is non-zero and non-unit. Hence $\langle a \rangle \neq R$. Let $\langle a \rangle \subseteq J \subseteq R$. Since J is also principal ideal $J = \langle b \rangle$ for some $b \in \mathbb{R}$. Now $a \in \langle a \rangle \subseteq \langle b \rangle \Rightarrow a = bx$ for some $x \in R$. Now since a is irreducible so either b or x is unit in R. If b is unit then $J = R$ on the other hand if x is unit then $\langle a \rangle = J$.
Conversely suppose that $\langle a \rangle$ is a maximal ideal. Let $a = bc$ for some $b, c \in R$. If neither b nor c is unit, then $\langle a \rangle$ is a proper subset of $\langle b \rangle$ and which is a proper subset of R which is a contradiction for $\langle a \rangle$ is maximal. □

8. *Let R be a PID. Then every irreducible element is prime.*

Proof. Let a be an irreducible element. Then $\langle a \rangle$ is maximal ideal hence prime ideal. Suppose $a|bc$, then $bc \in \langle a \rangle$. Then $b \in \langle a \rangle$ or $c \in \langle a \rangle$. Or equivalently $a|b$ or $a|c$. □

9. *Every nonzero prime ideal in PID is maximal.*

10. *Let R be PID and let $I_1 \subseteq I_2 \subseteq I_3 \subseteq I_4 \ldots$ be an ascending chain of ideals in a PID. Then there exists a $n \in \mathbb{Z}^+$ such that $I_n = I_{n+k}$ $\forall k \geq 1$.*

Proof. Since R is PID each I_i generated by a_i for some a_i. Then $I = \cup_{k=1}^{\infty} I_k$ is also an ideal in R. And is generated by single element say a. Then $a \in I_N$ for some N and it is easy to see that $I = I_N = I_{N+1} = I_{N+2} = \ldots$. □

11. *Show that every PID is UFD.*

Proof. Proof depends on the following results.

(a) $I_1 \subseteq I_2 \subseteq I_3 \subseteq I_4 \ldots$ be ideals in a ring R. Then $I = \cup_{k=1}^{\infty} I_k$ is also an ideal in R.

(b) Let $I_1 \subseteq I_2 \subseteq I_3 \subseteq I_4 \ldots$ be an ascending chain of ideals in a PID. Then there exists a $n \in \mathbb{Z}^+$ such that $I_n = I_{n+k}$ $\forall k \geq 1$.

Let a be non-zero and non-unit element of R. We prove that a has a irreducible factor. If a is irreducible, then a is irreducible factor to itself, and we are done. If a is reducible that is $a = a_1 b$, where neither a_1 nor b are zero and units, then $\langle a \rangle \subset \langle a_1 \rangle$. Further if a_1 is irreducible, then a_1 becomes an irreducible factor a. Otherwise we get ascending chain of ideals

$$\langle a \rangle \subset \langle a_1 \rangle \subset \langle a_2 \rangle \subset \langle a_3 \rangle \ldots$$

But in PID every ascending chain of ideals terminates. So there exists a_n for some n such that a_n is irreducible factor of a. Now $a = bc$ where b irreducible factor of a. If c is also irreducible then a has required factorization, otherwise c has irreducible factor which is also irreducible factor continuing in this way we show that a can written as product of irreducible factors.
uniqueness: Let $a = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$, where p_i, q_j are irreducible elements hence prime

elements. We need show $r = s$ and p_i is associate of q_j for some j. $p_1 | p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$. By re-labeling if necessary assume that $p_1 | q_1$, then p_1 and q_1 are associates, as ideals generated by them are maximal. By cancellation laws we get $p_2 p_2 \cdots p_r = u_1 q_2 q_3 \cdots q_s$. If $r < s$ then by repeating this processes after r steps we get $1 = u_1 2_2 u_3 \cdots u_r q_{r+1} \cdots q_s$. which is a contradiction hence $r = s$ and every p_i is associate to q_j for some j. $\square$

12. *Prove that in a PID two ideals $\langle a \rangle$ and $\langle b \rangle$ are co-maximal if and only if a greatest common divisor of a and b is 1 (in which case a and b are said to be coprime or relatively prime).*

13. *Let R be an integral domain and every nonzero prime ideal in $R[x]$ is maximal, then what can be say about R?*

14. *Find field fractions of $\mathbb{Z}, \mathbb{F}[x]$ and $\mathbb{F}[[x]]$, where $\mathbb{F}$ is a field.*

4 Lecture-4 (Euclidean and Polynomial rings)

Definition 4.1. *An integral domain R is said to be a* **Euclidean domain** *if for every every $a \neq 0$ in R there is defined a non-negative integer $N(a)$ such that*

1. *for all $a, b \in R$, both non-zero, $N(a) \leq N(ab)$.*
2. *for any $a, b \in R$, both non-zero, there exists $q, r \in R$ such that $a = bq + r$, where either $r = 0$ or $N(r) < N(b)$.*

Problem 4.2. 1. *Show that the set of integers is an Euclidean domain.*

Proof. We know $\mathbb{Z}$ is an integral domain.
Now the mapping $N : \mathbb{Z} \to \mathbb{Z}^+ \cup \{0\}$ defined as $N(x) = |x|$ further division algorithm in $\mathbb{Z}$ satisfies the required formalities. □

2. *Show that Euclidean domain is PID.*

Proof. Let R be an ED. And let I be an Ideal of R. If $I = \{0\}$, then I is principal ideal generated by 0, we are done.
Suppose $I \neq \{0\}$, then $0 \neq a \in I$ so that set $S = \{N(x) | 0 \neq x \in I\} \neq \emptyset$ hence by well ordering principle it has a least element say b.
Now we claim $I = \langle b \rangle$. Suppose $y \in I$, then since $b \neq 0$ by division algorithm there exists $q, r \in R$ such that $y = bq + r$, where $r = 0$ or $N(r) < N(b)$. Consequently, $r = 0$ follows from the fact that $r = y - bq \in I$ and $d(b)$ is the least in S. □

3. *Show that every field is Euclidean domain.*

Proof. Let $\mathbb{F}$ be a field. Define a mapping $N(x) = c \ \ \forall x \in \mathbb{F} \setminus \{0\}$, where $0 \leq c \in \mathbb{Z}$. It is clear that $N(ab) = N(a)$ and division algorithm is easily holds as $a = b(b^{-1}a) + 0$.
Further note from Problem 8 that this is the only way to define the mapping N. □

4. *Show that the set of Gaussian integers $\mathbb{Z}[i] = \{a + ib | a, b \in \mathbb{Z}\}$ is a Euclidean domain.*
5. *Let R be a Euclidean domain. Then any two elements a and b in R have a greatest common divisor d. Moreover $d = ax + by$ for some $x, y \in R$.*
6. *Let R be a Euclidean domain and $a, b \in R$. If $b \neq 0$ is not a unit in R, then $N(a) < N(ab)$.*
7. *Let R be a Euclidean domain. Suppose for $a, b, c \in R$, $a|bc$ but $(a, b) = 1$. Then $a|c$.*
8. *Prove that a necessary and sufficient condition that the element $a \neq 0$ in the Euclidean domain be a unit is that $N(a) = N(1)$.*

Proof. First note that if $0 \neq a \in R$, then $N(a) = N(a.1) \geq N(1)$. That is $N(a) \geq N(1) \forall a (\neq 0) \in R$
Suppose a is unit. Claim $N(a) = N(1)$.
We already have $N(1) \leq N(a)$. Now $N(1) = N(a.a^{-1}) \geq N(a)$.

Conversely, suppose $N(a) = N(1)$. Claim a is unit.
By division algorithm $1 = aq + r$, where either $r = 0$ of $N(r) < n(a) = N(1)$ but $N(1) \leq N(r)$ whenever $r \neq 0$. Hence $r = 0$.

□

9. *In a Euclidean domain prove that any two greatest common divisors of $a, b \in R$ are associates.*

Definition 4.3. *Let $\mathbb{F}$ be a field. By the ring of polynomials in the indeterminate, x, written as $\mathbb{F}[x]$, be the set of polynomials $a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$, where $n \in \mathbb{Z}^+ \cup \{0\}$ and $a_0, a_1, \ldots, a_n \in \mathbb{F}$.*

In order to make a ring out of $\mathbb{F}[x]$ we must able to recognize when two elements in it are equal. Further we must able to add and multiply elements of $\mathbb{F}[x]$.
If $a(x) = a_0 + a_1x + \cdots + a_mx^m$ and $b(x) = b_0 + b_1x + b_2x^2 + \cdots + b_nx^n$ are in $\mathbb{F}[x]$,

1. then $a(x) = b(x)$ if and only if for every integer $i \geq 0$, $a_i = b_i$.
2. $a(x) + b(x) = c_0 + c_1x + c_2x^2 + \cdots + c_tx^t$, where $c_i = a_i + b_i$ for all i.
3. $a(x).b(x) = c_0 + c_1x + c_2x^2 + \cdots + c_kx^k$, where $c_i = a_ib_0 + a_{i-1}b_1 + \cdots + a_1b_{i-1} + a_0b_i$ for all i.

Irreducible polynomial

Definition 4.4. *Let $f(x) = a_0 + a_1x + \cdots + a_nx^n \in \mathbb{F}[x]$ with $a_n \neq 0$. Then*

1. *the degree of $f(x)$ is denoted by $\deg(f(x))$ is n. We say that a polynomial is* **constant** *if its degree is zero. The degree of zero polynomial is defined as $-\infty$ (which is less than every real number).*
2. *$f(x)$ is said to be* monic *if $a_n = 1$.*
3. *$f(x)$ is said to divide $g(x)$ in $\mathbb{F}[x]$ if there exists $h(x) \in \mathbb{F}[x]$ such that $g(x) = f(x)h(x)$.*
4. *An element $f(x) \in \mathbb{F}[x]$ is said to be* **irreducible polynomial over $\mathbb{F}$** *if "whenever $f(x) = g(x)h(x)$, with $g(x), h(x) \in \mathbb{F}[x]$ then either $\deg(g(x)) = 0$ or $\deg(h(x)) = 0$" that is one of $g(x)$ or $h(x)$ is a constant polynomial.*

With the definitions as above, it can be easily observed that *division algorithm* holds in $\mathbb{F}[x]$. That is, if $f(x), g(x) \in \mathbb{F}[x]$ with $\deg(g(x)) > 1$ then there exist $q(x), r(x) \in \mathbb{F}[x]$ such that $f(x) = q(x)g(x) + r(x)$, with either $r(x) \equiv 0$ or $\deg(r(x)) < \deg(g(x))$. One now defines, the *greatest common divisor* of two elements $g(x), h(x) \in \mathbb{F}[x]$, denoted $\gcd(g(x), h(x))$, as a monic polynomial $d(x) \in \mathbb{F}[x]$ such that

1. $d(x)$ divides both $g(x)$ and $h(x)$ in $\mathbb{F}[x]$, and
2. if there is a polynomial $f(x) \in \mathbb{F}[x]$ that divides both $g(x)$ and $h(x)$ in $\mathbb{F}[x]$ then $f(x)$ divides $d(x)$ in $\mathbb{F}[x]$.

Then using the division algorithm one proves that $d(x) = \gcd(g(x), h(x))$ exists, for any two polynomials $g(x), h(x) \in \mathbb{F}[x]$. Also, one proves the *Euclid's algorithm* which states that there exist polynomials $a(x), b(x) \in \mathbb{F}[x]$ such that $d(x) = \gcd(g(x), h(x)) = a(x)g(x) + b(x)h(x)$. The above results can be used to prove that if I is a proper ideal in $\mathbb{F}[x]$ then there exists $d(x) \in \mathbb{F}[x]$ such that $I = \langle d(x) \rangle = \{d(x)g(x) : g(x) \in \mathbb{F}[x]\}$.

Problem 4.5. *1. Show that $\mathbb{F}[x]$ is Euclidean domain, where $\mathbb{F}$ is a field.*

2. If $f(x), g(x) \in \mathbb{F}[x]$ are two non-zero elements (polynomials) in $\mathbb{F}[x]$, then

(a) $\deg(f(x)g(x)) = \deg(f(x)) + \deg(g(x))$.

(b) $\deg(f(x)) \leq \deg(f(x)g(x))$.

3. Show that $\mathbb{Z}[\sqrt{-5}]$ is not a UFD.

Proof. First note that $\mathbb{Z}[\sqrt{-5}] = \{a+b\sqrt{-5} | a, b \in \mathbb{Z}\}$. We define a mapping $N : \mathbb{Z}[\sqrt{-5}] \to \mathbb{Z}^+ \cup \{0\}$ as

$$N(x + y\sqrt{-5}) = (x + y\sqrt{-5})(x - y\sqrt{-5}) = x^2 + 5y^2.$$

It is easy to see that the function N is multiplicative that is $N(z_1 \cdot z_2) = N(z_1)N(z_2)$.

$\boxed{\text{In } \mathbb{Z}[\sqrt{-m}] \text{ we have } (a^2 + mb^2)(c^2 + md^2) = (ac - mbd)^2 + m(ad + bc)^2, \text{ where } m \text{ is a positive integer.}}$

If $x + y\sqrt{-5}$ be a unit in $\mathbb{Z}[\sqrt{-5}]$, then there exist $a + b\sqrt{-5} \in \mathbb{Z}[\sqrt{-5}]$ such that

$$(x + y\sqrt{-5})(a + b\sqrt{-5}) = 1.$$

So $x^2 + 5y^2 = 1 = a^2 + 5b^2$ this is possible only when $x = \pm 1, y = 0$ consequently only units in $\mathbb{Z}[\sqrt{-5}]$ are ± 1. Now we observe that

$$6 = 2.3 = (1 + \sqrt{-5})(1 - \sqrt{-5}).$$

It also clear that $2, 3, 1 + \sqrt{-5}, 1 - \sqrt{-5}$ are not units in $\mathbb{Z}[\sqrt{-5}]$. Now we claim that

(a) $2, 3, 1 + \sqrt{-5}, 1 - \sqrt{-5}$ are irreducible.

(b) $2, 3, 1 + \sqrt{-5}, 1 - \sqrt{-5}$ are not associates. This one is easy to see, because only units in $\mathbb{Z}[\sqrt{-5}]$ are ± 1. Alternative way to see this is $N(2), N(3)$ and $N(1 + \sqrt{-5})$ are distinct.

For clarification note that from the factorization of 48 as $4 \cdot 12$ and $6 \cdot 8$, we should not claim that $\mathbb{Z}$ is not UFD. As $4, 12, 6, 8$ are reducible elements in $\mathbb{Z}$.
First we prove 2 is irreducible. On the contrary suppose 2 is reducible. That is $2 = (x + y\sqrt{-5})(a + b\sqrt{-5})$, where $(x + y\sqrt{-5}), (a + b\sqrt{-5})$ are non-zero and non-units. But we have

$$N(2) = N((x + y\sqrt{-5})N(a + b\sqrt{-5})$$

$$4 = (x^2 + 5y^2)(a^2 + 5b^2).$$

Consequently, $x^2 + 5y^2 = 2$ but this does not have any integer solution. Hence our assumption 2 is reducible is wrong. In the similar fashion we can show that $3, 1 + \sqrt{-5}, 1 - \sqrt{-5}$ are irreducible. $\square$

4. Show that $\mathbb{Z}[\sqrt{-3}]$ is not a UFD. But $\mathbb{Z}[\frac{1+\sqrt{-3}}{2}] = \mathbb{Z}[\omega]$, is UFD, where $\omega = \frac{1+\sqrt{-3}}{2}$, the cube root of unity and the set $\mathbb{Z}[\frac{1+\sqrt{-3}}{2}]$ is called set of **Eisensteinian integers**.

Proof. First note that $\mathbb{Z}[\sqrt{-3}] = \{a+b\sqrt{-3} | a, b \in \mathbb{Z}\}$. We define a mapping $N : \mathbb{Z}[\sqrt{-3}] \to \mathbb{Z}^+ \cup \{0\}$ as

$$N(x + y\sqrt{-3}) = (x + y\sqrt{-3})(x - y\sqrt{-3}) = x^2 + 3y^2.$$

If $x + y\sqrt{-3}$ be a unit in $\mathbb{Z}[\sqrt{-3}]$, then there exist $a + b\sqrt{-3} \in \mathbb{Z}[\sqrt{-3}]$ such that

$$(x + y\sqrt{-3})(a + b\sqrt{-3}) = 1.$$

So $x^2 + 3y^2 = 1 = a^2 + 3b^2$ this is possible only when $x = \pm 1, y = 0$ consequently only units in $\mathbb{Z}[\sqrt{-3}]$ are ± 1. Now we observe that

$$4 = 2.2 = (1 + \sqrt{-3})(1 - \sqrt{-3}).$$

It also clear that $2, 1 + \sqrt{-3}, 1 - \sqrt{-3}$ are not units in $\mathbb{Z}[\sqrt{-3}]$. Now we claim that

(a) $2, 1 + \sqrt{-3}, 1 - \sqrt{-3}$ are irreducible.

(b) $2, 1 + \sqrt{-3}, 1 - \sqrt{-3}$ are not associates. This one is easy to see, because only units in $\mathbb{Z}[\sqrt{-3}]$ are ± 1. Alternative way to see this is $N(2), N(1 + \sqrt{3})$ are distinct.

First we prove 2 is irreducible. On the contrary suppose 2 is reducible. That is $2 = (x + y\sqrt{-3})(a + b\sqrt{-3})$, where $(x + y\sqrt{-3}), (a + b\sqrt{-3})$ are non-zero and non-units. But we have

$$N(2) = N((x + y\sqrt{-3})N(a + b\sqrt{-3})$$

$$4 = (x^2 + 3y^2)(a^2 + 3b^2).$$

Consequently, $x^2 + 3y^2 = 2$ but this does not have any integer solution. Hence our assumption 2 is reducible is wrong. In the similar fashion we can show that $1 + \sqrt{-3}, 1 - \sqrt{-3}$ are irreducible. But once we work in $\mathbb{Z}[\omega]$ which contains $\mathbb{Z}[\sqrt{-3}]$. One can see that $\omega \cdot \omega^2 = 1$. That is ω is a unit in $\mathbb{Z}[\omega]$. Consequently, $2 \cdot \omega = 2 \cdot \frac{1+\sqrt{-3}}{2} = 1 + \sqrt{-3}$ that is 2 and $1 + \sqrt{-3}$ are associates in $\mathbb{Z}[\omega]$. □

5. *Find all units in the ring* $\mathbb{Z}[i]$.

Proof. Let $a + bi \in \mathbb{Z}[i]$ be a unit. Then $a^2 + b^2 = 1$ it is possible only if $a = \pm 1, b = 0$ or $a = 0, b = \pm 1$. Thus $U(\mathbb{Z}[i]) = \{1, -1, i, -i\}$. □

6. *Find all units in the ring* $\mathbb{Z}[\omega]$.

Proof. First note that

(a) ω is root of $x^3 - 1$ in particular root of $x^2 + x + 1$. So $\omega^3 = 1 \Rightarrow \omega.\omega^2 = 1$ also $|\omega|^2 = \omega \cdot \overline{\omega} = 1$ Hence $\omega^2 = \overline{\omega}$.

(b) if $\alpha = a + b\omega \in \mathbb{Z}[\omega]$, then $\overline{\alpha}$, the conjugate of α is defined as $a + b\overline{\omega} = a + b\omega^2$.

(c) $N(\alpha) = N(a + b\omega) = (a + b\omega)(a + b\omega^2) = a^2 + ab[\omega + \omega^2] + b^2\omega^3 = a^2 - ab + b^2$.

Suppose $\alpha = a + b\omega \in \mathbb{Z}[\omega]$ be a unit. Then $a^2 - ab + b^2 = 1$ or equivalently $4(a^2 - ab + b^2) = 4$. That is $(2a - b)^2 + 3b^2 = 4$. This is possible if

(a) $2a - b = \pm 1, b = \pm 1$ or

(b) $2a - b = \pm 2, b = 0$

If 2a-b=1,b=1, then $\alpha = 1 + \omega$.

If 2a-b=1,b=-1, then $\alpha = -\omega$.

If 2a-b=-1,b=1, then $\alpha = \omega$.

If 2a-b=-1,b=-1, then $\alpha = -1 - \omega$.

If 2a-b=$\pm$ 2, b= 0, then $\alpha = 1, -1$ respectively.

So $U(\mathbb{Z}[\omega]) = \{1, -1, \omega, -\omega, \omega^2, -\omega^2\}$. □

7. *Let $\pi \in \mathbb{Z}[i]$ and $N(\pi) = p$, where p is a rational prime. Then π is a Gaussian prime. (Prime numbers in $\mathbb{Z}$, the set of integers are called rational primes to distinguish them from primes in $\mathbb{Z}[i]$, the Gaussian integers, primes in $\mathbb{Z}[\omega]$ the Eisenstein integers.)*

   Proof. Let $\pi = \gamma\delta$ for some $\gamma, \delta \in \mathbb{Z}[i]$.

$$p = N(\pi) = N(\gamma)N(\delta).$$

   Hence at least one of $N(\gamma)$ and $N(\delta)$ is 1 or equivalently either γ or δ is a unit. Thus π is an irreducible hence prime element. □

8. *Every prime in $\mathbb{Z}[i]$ divides exactly one rational prime.*

   Proof. Let π be a Gaussian prime and $\pi | p$, where p be a rational prime (prime in $\mathbb{Z}$). Let p, q be distinct primes in $\mathbb{Z}$, then $\gcd(p, q) = 1$ or equivalently there exists $x, y \in \mathbb{Z}$ such that $px + qy = 1$. So if $\pi | q$, then $\pi | 1$, which is a not possible. □

9. *Find all primes in $\mathbb{Z}[i]$ which divide 2.*

   Proof. Since $2 = 1^2 + 1^2 = (1 + i)(1 - i)$. Further $1 - i = -i(1 + i)$ so $2 = -i(1 + i)^2$ and $N(1+i) = 2$, hence $1+i$ is a prime. Therefore by unique factorization in $\mathbb{Z}[i]$ $1+i$ and its associates $-1 - i, -1 + i, 1 - i$ are the only primes divide 2. □

10. *Let p be a rational prime. Then p factors over $\mathbb{Z}[i]$ as follows*

    (a) *If $p = 2$, then $2 = -i(1 + i)^2$ and $N(1 + i) = 2$.*

    (b) *If $p \equiv 3 \pmod 4$, then $p = \pi$ be a Gaussian prime and $N(\pi) = p^2$.*

    Proof. If $p = \gamma\delta$, then $p^2 = N(\gamma)N(\delta)$. Consequently $N(\gamma) = 1$ *or* p *or* p^2. We claim that $N(\gamma) \neq p$. Suppose $\gamma = a + ib$ and $p = N(\gamma) = a^2 + b^2$. Then clearly a and b are of different parity so $a^2 + b^2$ not congruent to $3 \pmod 4$. Hence either γ or β is unit in $\mathbb{Z}[i]$. □

    (c) *If $p \equiv 1 \pmod 4$, then $p = \alpha\beta$, where α, β are non-associate primes, $N(\alpha) = N(\beta) = p$. Further α, β are unique up to order and associates.*

    Proof. If $p \equiv 1 \pmod 4$, then it is known that $p = x^2 + y^2$ for some $x, y \in \mathbb{Z}$. Hence $p = (x + iy)(x - iy)$, and it is to check $(x + iy)$ and $(x - iy)$ are not associates. The associates of $x + iy$ are $\{x + iy, -x - iy, ix - y, -ix + y\}$. □

5 Lecture-5,Polynomials over the Rational Field

Definition 5.1. 1. *The* **content** *of the polynomial* $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$, *where the coefficients* $a_0, a_1, \ldots, a_n$ *are integers, denoted by* $c(f)$ *is the greatest common divisor of the integers* $a_0, a_1, \ldots, a_n$.

2. *The polynomial* $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$, *where the coefficients* $a_0, a_1, \ldots, a_n$ *are integers is said to be* **primitive** *if the greatest common divisor of* $\{a_0, a_1, \ldots, a_n\}$ *is* 1 *(or its content* $c(f)$ *is* 1.*)*

Problem 5.2. *Following results are applied to any unique factorization domain* R *with* $\mathbb{F}$ *as its field of fractions.*

1. *(Gauss Lemma) If* $f(x)$ *and* $g(x)$ *are primitive polynomials, then* $f(x)g(x)$ *is a primitive polynomial.*

   Proof. Let $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ and $g(x) = b_0 + b_1x + b_2x^2 + \cdots + b_mx^m$ be primitive polynomials in $\mathbb{Z}[x]$. Suppose $f(x)g(x)$ is not primitive. That is there exists a prime $p \in \mathbb{Z}$ such that $p|c(fg)$. Since $p \nmid c(f), c(g)$ there exists smallest r, s with $0 \leq r \leq n$ and $0 \leq s \leq m$ such that $p \nmid a_r$ and $p \nmid b_s$ but $p|a_0, a_1, \ldots a_{r-1}, b_0, b_1, \ldots, b_{s-1}$. Now consider the coefficient $a_0b_{r+s} + a_1b_{r+s-1} + \cdots + a_{r-1}b_{s+1} + a_rb_s + a_{r+1}b_{s-1} + \ldots a_{r+s}b_0$ of x^{r+s} in $f(x)g(x)$ which is divisible by p.
   (Note that $a_i, b_j = 0$ if $i > n$ and $j > m$ respectively.)
   Also $p|a_0b_{r+s} + a_1b_{r+s-1} + \cdots + a_{r-1}b_{s+1} + a_{r+1}b_{s-1} + \ldots a_{r+s}b_0$, hence $p|a_rb_s$. We got required contradiction hence $f(x)g(x)$ is also primitive.
   Alternate proof is reduce to $\mathbb{Z}_p$ and use the fact that $\mathbb{Z}_p[x]$ is ID. □

2. *Let* R *be a UFD and* $\mathbb{F}$ *be its field of fractions. Let* $f(x) \in R[x]$ *be a polynomial of degree at least one. Then* $f(x)$ *is irreducible in* $R[x]$ *if and only if* $f(x)$ *is primitive in* $R[x]$ *and* $f(x)$ *is irreducible in* $\mathbb{F}[x]$.

   Proof. $\Rightarrow$ If $f(x) \in R[x]$ is irreducible, then it is easy to that f is primitive. If $f(x)$ is reducible in $\mathbb{F}[x]$, then $f(x) = g_1(x)h_1(x)$, where $g_1(x), h_1(x) \in \mathbb{F}[x]$, hence there exists $a, b, c, d \in R$ such that $abf(x) = cdg(x)h(x)$ and $g(x), h(x) \in R[x]$ and primitive. Hence $\frac{cd}{ab}$ is unit in R. That is $f(x)$ is reducible in $R[x]$.

   $\Leftarrow$ Suppose that $f(x)$ is primitive in $R[x]$ and $f(x)$ irreducible in $\mathbb{F}[x]$. We need to show that $f(x)$ is irreducible in $R[x]$. Suppose $f(x) = g(x)h(x)$ in $R[x]$. Then $f(x) = g(x)h(x)$ in $\mathbb{F}[x]$. But $f(x)$ is irreducible in $\mathbb{F}[x]$. Either $g(x)$ or $h(x)$ is a unit in $\mathbb{F}[x]$. But both $g(x)$, $h(x)$ are in $R[x]$. Hence one of them is unit in $R[x]$ follows from the fact that $f(x)$ is primitive.

   □

3. *If the primitive polynomial* $f(x)$ *can be factored as the product of two polynomials having rational coefficients, it can be factored as the product of two polynomials having integer coefficients.*

4. *If an integer monic polynomial factors as the product of two non-constant polynomials having rational coefficients then it factors as the product of two integer monic polynomials.*

5. *A non-constant polynomial in $\mathbb{Z}[x]$ is irreducible in $\mathbb{Z}[x]$ if and only if it is both irreducible in $\mathbb{Q}[x]$ and primitive in $\mathbb{Z}[x]$.*

6. **(The Eisenstein Criterion)** *Let $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ be a polynomial with integer coefficients. Suppose that for some prime number p, $p \nmid a_n, p|a_1, p|a_2, \ldots, p|a_0, p^2 \nmid a_0$. Then $f(x)$ is irreducible over the rationals. Further if $f(x)$ is primitive then it irreducible in $\mathbb{Z}[x]$.*

   Proof. Let $f(x) = c(f)f_1(x)$, where $c(f)$ is the content of f which is a unit in $\mathbb{Q}$ hence in $\mathbb{Q}[x]$. Thus in order to show f is irreducible in $\mathbb{Q}[x]$, it is sufficient to show $f_1(x)$ is irreducible in $\mathbb{Q}[x]$. On the contrary suppose that $f_1(x) = g(x)h(x)$, where $g(x) = b_0 + b_1x + b_2x^2 + \cdots + b_kx^k$, $h(x) = c_0 + c_1x + c_2x^2 + \cdots + c_tx^t$, with $1 \leq k, t < n$. Note $p \nmid a_n$, so $p \nmid c(f)$, hence the coefficients of $f_1 = a_0^* + a_1^*x + a_2^*x^2 + \cdots + a_n^*x^n$ satisfy the same divisibility conditions with respect to p as they do with coefficients of f. Now since $p|a_0^* = b_0c_0$ so p divides b_0 or c_0. With out loss of generality assume that $p|b_0$. So $p \nmid c_0$ follows from the fact that $p^2 \nmid a_0^*$. From the primitivity of f_1, p does not divide all the coefficients of $g(x)$, suppose s is the least positive (as $p|b_0$) such that $p \nmid b_s$. That is $p|b_0, b_1, \ldots, b_{s-1}$ and $p \nmid b_s$. Consider a_s* since $a_s^* = b_0c_s + b_1c_{s-1} + \cdots + b_sc_0$ and $p|a_s^*$ as $s \leq k < n$ and $p|b_0, b_1, \ldots, b_{s-1}$ forces $p|b_sc_0$ which is false. Hence $f_1(x)$ is irreducible.

   □

7. *Let $a \in \mathbb{F}, f(x) \in \mathbb{F}[x]$. Then $x - a$ divides $f(x) \in \mathbb{F}[x]$ if and only if $f(a) = 0$.*

   Proof. First note that $\mathbb{F}[x]$ is an ED. So by division algorithm, we have $f(x) = (x - a)q(x) + r(x)$, where $r(x) = 0$ or $\deg(r(x)) < \deg((x - a)) = 1$.
   Or equivalently $f(x) = (x - a)q(x) + r$. Further $r = f(a)$ as $f(a) = (a - a)q(a) + r$.
   Thus $x - a$ divides $f(x)$ if and only if $f(a) = 0$. □

8. *If R is a commutative ring and $f(x) \in R[x]$ and $f(\alpha) = 0$ for some $\alpha \in R$. Then $x - \alpha|f(x)$.*

   Proof. Let $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$. Then

   $$f(x) = f(x) - f(\alpha) = a_1(x - \alpha) + a_2(x^2 - \alpha^2) + \cdots + a_n(x^n - \alpha^n).$$

   It is easy to check that $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \cdots + ab^{n-2} + b^{n-1})$. Hence the result follows. □

9. *Let R be a ring. Then $f(x) = ax + b$ has a root in R if and only if $a|b$.*

   Proof. If $\alpha \in R$ is a root of $f(x)$, then $b = a(-\alpha)$. Conversely suppose that $b = ac$ for some $c \in R$. Then

   $$ax + b = ax + ac = a(x + c).$$

   Easy to check that $-c \in R$ is a root of $f(x)$. □

10. *If $R = \mathbb{Z}_6$, then $f(x) = 2x$ has two roots 0 and 3.*

11. *If R is an integral domain, then $f(x) = ax + b$ has at most one root.*

Proof. If $f(x)$ has no roots, then nothing to prove. Suppose α and β are two distinct roots, then $a(\alpha - \beta) = 0$ since $a \neq 0$, $\alpha = \beta$. $\square$

12. *If R is an integral domain, then a polynomial of degree n has at most n roots.*

Proof. By induction on n. $\square$

13. *If $f(x) \in \mathbb{F}[x]$ and degree of $f(x)$ is either 2 or 3, then $f(x)$ is reducible if and only if there exists $a \in \mathbb{F}$ such that $f(a) = 0$.*

Proof. Let $f(x)$ be a reducible polynomial of degree 2 or 3. That is $f(x) = g(x)h(x)$, where $h(x), g(x)$ are neither zero nor units. We also know that every non-zero constant polynomial is unit. Consequently, $\deg(g(x)), \deg(h(x)) \geq 1$. Finally, result follows from the fact that $\deg(f(x) = \deg(g(x)) + \deg(h(x))$. $\square$

14. *The ideal $A = \langle p(x) \rangle$ in $\mathbb{F}[x]$ is a maximal ideal if and only if $p(x)$ is irreducible over the field $\mathbb{F}$.*

15. *Show that $\mathbb{R}[x]/\langle x^2 + 1 \rangle \cong \mathbb{C}$.*

Proof. First note that x^2+1 has no roots in $\mathbb{R}$. Thus it does not factor as linear polynomials in $\mathbb{R}[x]$. Now since $\mathbb{R}[x]$ is PID and x^2+1 is an irreducible element in $\mathbb{R}[x]$, hence $\langle x^2+1 \rangle$ is a maximal ideal. Or equivalently $\mathbb{R}[x]/\langle x^2+1 \rangle$ is field. It is easy to see that $\mathbb{R}[x]/\langle x^2+1 \rangle = \{a + bx \pmod{\langle x^2+1 \rangle}\}$. Now we can define a mapping $\phi : \mathbb{R}[x]/\langle x^2 + 1 \rangle \to \mathbb{C}$ as $\phi(a + bx \pmod{\langle x^2 + 1 \rangle}) = a + bi$ which is an isomorphism.

Alternate proof

Define a mapping $f : \mathbb{R}[x] \to \mathbb{C}$ as $f(p(x)) = p(i)$ that is evaluation at i. One can easily check that f is ring homomorphism and on-to.

Homomorphism: $f(a(x) + b(x)) = a(i) + b(i) = f(a(x)) + f(b(x)) f(a(x)b(x)) = a(i)b(i) = f(a(x))f(b(x))$.

Onto: Let $\alpha = a + bi \in \mathbb{C}$. Then $f(a + bx) = \alpha$.

Kernal of f: Let $p(x) \in Kerf$. Then $p(i) = 0$. Now by division algorithm in $\mathbb{R}[x]$ we have $p(x) = (x^2 + 1)q(x) + r(x)$ and $p(i) = 0$ forces $r(x) = 0$, that $p(x) \in \langle x^2 + 1 \rangle$. Further it is easy to see that every element in $\langle x^2 + 1 \rangle$ is in $Kerf$.

Hence by first Fundamental theorem of homomorphism $\mathbb{R}[x]/\langle x^2 + 1 \rangle \cong \mathbb{C}$.

$\square$

16. *Find all ideals in the ring $\mathbb{F}[x]/\langle p(x) \rangle$, where $\mathbb{F}$ is field and $p(x) \in \mathbb{F}[x]$*

Theorem 5.3 (The remainder Theorem)**.** *Let $\mathbb{F}$ be a field, $a \in \mathbb{F}$, and $f(x) \in \mathbb{F}[x]$. Then $f(a)$ is the remainder in the division of $f(x)$ by $x - a$.*

Theorem 5.4 (The Factor Theorem)**.** *Let $\mathbb{F}$ be a field, $a \in \mathbb{F}$, and $f(x) \in \mathbb{F}[x]$. Then a is a zero of $f(x)$ if and only if $(x-a)$ is a factor of $f(x)$*

Theorem 5.5. *A polynomial of degree n over a field has at most n zeros counting multiplicity.*

Theorem 5.6 (Reducibility Test for Degrees 2 and 3)**.** *Let $\mathbb{F}$ be a field. If $f(x) \in \mathbb{F}[x]$ and $\deg\ f(x) = 2$ or 3, then $f(x)$ is reducible over $\mathbb{F}$ if and only if $f(x)$ has a zero in $\mathbb{F}$.*

Theorem 5.7 (Mod p irreducibility Test)**.** *Let p be a prime and suppose $f(x) \in \mathbb{Z}[x]$ with $\deg(f(x) \geq 1$. Let $\overline{f}(x)$ be a polynomial in $\mathbb{Z}_p[x]$ obtained from $f(x)$ by reducing all the coefficients of $f(x)$ modulo p. If $\overline{f}(x)$ is irreducible over $\mathbb{Z}_p$ and $\deg\ \overline{f}(x) = \deg\ f(x)$, then $f(x)$ is irreducible over $\mathbb{Q}$.*

Problem 5.8.

1. *Show that x^4+1 is irreducible over $\mathbb{Q}$ but reducible over $\mathbb{R}$.*
2. *Show that x^2+x+4 is irreducible over $\mathbb{Z}_{11}$.*
3. *Show that a polynomial from $\mathbb{Z}[x]$ that is irreducible over $\mathbb{Z}$ is primitive.*
4. *Show that x^4+1 is reducible over $\mathbb{Z}_p$ for every prime p.*
5. *Let p be a prime. Then show that the number of reducible polynomial over $\mathbb{Z}_p$ of the form x^2+ax+b is $\frac{p(p+1)}{2}$. Also find the number of irreducible polynomials $\mathbb{Z}_p$ of the form x^2+ax+b*
6. *Write $f(x) = x^3+6 \in \mathbb{Z}_7[x]$ as product of irreducible polynomials over $\mathbb{Z}_7$.*
7. *Find all monic polynomials of degree 2 in $\mathbb{Z}_3$.*
8. *Find all factors of x^n-1 over $\mathbb{Z}[x]$.*

Similarities between $\mathbb{Z}$ and $\mathbb{F}[x]$.

	$\mathbb{Z}$	$\mathbb{F}[x]$
1	Form of elements $a_n 10^n + a_{n-1}10^{n-1} + \cdots + a_1 10 + a_0$	Form of elements $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$
2	Euclidean Domain $d(a) = \lvert a \rvert$	Euclidean Domain $d(f(x)) = \deg\ f(x)$
3	Units: a is unit if and only if $\lvert a \rvert = 1$	Units: $f(x)$ is unit if and only if $\deg\ f(x) = 0$.
4	Division algorithm For $a, b \in \mathbb{Z}$, $b \neq 0$, there exists $q, r \in \mathbb{Z}$ such that $a = bq + r, 0 \leq r < \lvert b \rvert$	Division algorithm: For $f(x), g(x) \in \mathbb{F}[x]$, $g(x) \neq 0$, there exists $q(x), r(x) \in \mathbb{F}[x]$ such that $f(x) = g(x)q(x) + r(x), 0 \leq \deg\ r(x) < \deg\ g(x)$ or $r(x) = 0$.
5	PID Every ideal $I = \langle a \rangle$ where $\lvert a \rvert$ is minimum	PID:Every ideal $I = \langle f(x) \rangle$, where $\deg\ f(x)$ is minimum ($I \neq \{0\}$)
6	Prime number	Irreducible polynomial
7	UFD Every element is a "unique" product of primes	UFD Every element is a "unique" product of irreducibles

5.1 More problems for the Tutorials

Definition 5.9. *Let R be a ring and $a \in R$. Then annihilator of a denoted $ann(a)$ and is defined as $ann(a) = \{x \in R | ax = 0\}$.*

Theorem 5.10. *Let R be a commutative ring with unity.*

1. *If $a \in R$, then $ann(a)$ is an ideal of R.*
2. *If we define a relation on R as $a \sim b$ if and only if $ann(a) = ann(b)$, then $\sim$ is an equivalence relation on R. In particular if R is an integral domain if and only if $ann(x) = \{0\}$ for any non zero element $x \in R$. Hence there are two equivalence classes. It parallel to associates in fields.*

Let R be a commutative ring with unity and $a \in R \setminus \{0\}$. Then find a map from R to R.

- $a \mapsto a + r$
- $a \mapsto a + r$

Proposition 5.11. *Let R be a finite commutative ring with unity. Then a non zero element is either a unit or a zero divisor. Hence $R = Z(R) \cup U(R)$.*

Proof.
- Let $R = \{a_1, a_2, \ldots, a_n\}$.
- Let $a \in R$ be a non zero, non unit and $S = \{aa_i | a_i \in R\}$.
- Then $1 \notin S$ and there exists $i \neq j$, such that

  $aa_i = aa_j$ That is $a(a_i - a_j) = 0$. Hence a is a zero divisor.

Alternate proof First note that every finite commutative ring Artinian (invited to prove). Let x be an element of an Artinian ring A and consider the sequence of ideals

$$Ax \supset Ax^2 \supset Ax^3 \supset \ldots$$

By the descending chain condition this sequence of ideals must become constant after some stage. In other words, there must be an element y in A such that $yx^{k+1} = x^k$ for some (sufficiently large) integer k. Given such a y, let k be the smallest non-negative integer for which this equation holds (with the convention that $x^0 = 1$). It is clear that if $yx - 1 = 0$ then x is a unit; in other words x is an element of A^*, the group of units of A. On the other hand if $yx - 1 \neq 0$ then we see that $k > 0$ and $(yx - 1)x^{k-1} \neq 0$; since $(yx - 1)x^{k-1} \cdot x = 0$, we see that x is a zero divisor. Thus every element of A is either a unit or a zero-divisor. □

- Let R be a commutative ring with unity and $R \neq 0$. Then TFAE.
  - R is field.
  - The only ideals in R are $\langle 0 \rangle$ and R.
  - every nonzero homomorphism from R to a nonzero ring S is injective.

Proof. Suppose R is field and $I \neq \{0\}$ be an ideal of R, then $0 \neq x \in I$ then $1 = x^{-1}x \in I$. Thus $I = R$.

Suppose only ideals in R are $\langle 0 \rangle$ and R. Let $f : R \to S$ be homomorphism, then $ker\ f$ is an ideal of R. If $ker\ f = R$, then f is trivial (or zero) homomorphism. Else ker $f = \{0\}$.

Let $x \in R$ is not a unit. Let $B = A/\langle x \rangle$. Then B is nonzero ring. Define a map f from A to B as $f(a) = a + \langle x \rangle$, then it is easy check that f is nonzero homomorphism with kernal $\langle x \rangle$. From hypothesis f is injective hence $\langle x \rangle = \{0\}$. □

- Let R be a commutative ring with unity. Then

  1. M is maximal ideal if and only if R/M is field.
  2. M is prime ideal if and only if R/M is an integral domain. (A zero ideal is prime if only if R is ID.)
  3. every maximal ideal is prime.

  Proof. Proof of Part 1: Suppose M is maximal ideal. Claim R/M is field. Let $a+M$ be nonzero element in R/M. That is $a + M \neq M$ or $a \notin M$. Then $\langle a \rangle + M = R$. That is there exists $b \in R$, $x \in M$ such that $ab + x = 1$. Thus $ab + x + M = 1 + M$ or $(a + M)(b + M) = 1 + M$.

  Conversely suppose that R/M is field. Suppose $M \subsetneq J \subseteq R$. Let $a \in J \setminus M$, then $a + M$ is a unit in R/M. There exists $b \in R$ such that $(a + M)(b + M) = 1 + M$ that $ab - 1 \in M \subset J$. Since $a \in J$ hence $1 \in J$ or $J = R$. □

- Zorn's Lemma: Let S be a partially ordered set. If every totally ordered subset of S has an upper bound, then S contains a maximal element.

- Every nonzero commutative ring with unity contains a maximal ideal.

  Proof.
  - Let S be the set of proper ideals in a commutative ring $R \neq 0$. Since the zero ideal $\{0\}$ is a proper ideal, $S \neq \emptyset$. We partially order S by inclusion.
  - Let $\{I_a\}_{a \in A}$ be a totally ordered set of proper ideals in R. To write down an upper bound for these ideals in S, it is natural to try their union $I = \cup_{a \in A} I_a$.
  - Easy to show that I is an ideal, contains I_a for every $a \in A$ and is proper (here we use R is a ring with unity, suppose I is not proper, then $I = R$ hence $1 \in I$ and $1 \in I_a$ from $a \in \mathcal{A}$, but I_a is proper), hence belongs to S. I is upper bound for totally ordered set $\{I_a\}_{a \in A}$.
  - By Zorn's lemma S has maximal element say J. Now we claim J is maximal ideal. If J is not maximal ideal, then there exists a proper ideal say K in R such that $J \subsetneq K$. Then $K \in S$ and which contradicts the maximality of J.

  □

- As a corollary to above result we have

  - Let I be an ideal of $R \neq I$.Then there is a maximal ideal containing I.
    Proof is same as above S is set of proper ideals of R which contains I. Another way to prove is considering A/I. Since I is proper ideal hence $A/I \neq \{0\}$. Then A/I has a maximal ideal. Result follows easily from this.

- Every non unit of a nonzero ring R is contained in some maximal ideal.
  Consider the principal ideal generated by a non unit.

- Let R be a commutative ring with unity and $I = \{a \in R | a \text{ is not unit}\}$. Then I is a proper nonempty set, as $0 \in I, 1 \notin I$. Is I an ideal? Clearly the answer is no as if $R = \mathbb{Z}$, and $I = \mathbb{Z} \setminus \{-1, 1\}$ is not an ideal in $\mathbb{Z}$. What happens if it is an ideal? Is it maximal?

  Proof. If $x \in I$, then there exists a maximal ideal M containing x. Now if $y \in M$, then y is not a unit hence $y \in I$. That is $M \subseteq I \subsetneq R$. Hence $I = M$ and I is maximal. In fact we proved more any maximal ideal is I. Equivalently I is unique maximal ideal in R. □

  Check $\mathbb{Z}/p^n\mathbb{Z}$ where p is prime will work?. What are the interesting subsets of set of non-units?. Yes. The set of zero divisors and set of nilpotent elements. Find all nilpotent elements in $\mathbb{Z}/30\mathbb{Z}$.

Let A be a commutative ring. The set of all nilpotent elements of A (i.e., the set of all x for which $x^n = 0$ for some natural n) is called the nilradical of A. It is sometimes denoted $\mathfrak{N}$.

The nilradical is an ideal, for if $x^n = 0$, then for any $a \in A$, $(ax)^n = 0$; and if $x^n = y^m = 0$, then $(xy)^{mn} = 0$,

$$(x+y)^{m+n-1} = \sum_{k=0}^{m+n-1} \binom{n}{k} x^{m+n-1-k} y^k = 0.$$

Hence $\mathfrak{N}$ is a proper ideal of R.

Proposition 5.12. *The nilradical $\mathfrak{N}$ of A is the intersection of all prime ideals of A.*
(The intersection of all prime ideals in a commutative ring is the set of nilpotent elements in the ring.)

Proof. (This is striking: it tells us that if we know an element in a commutative ring lies in every prime ideal, some power of it must be 0)

Let $\mathfrak{p}$ be a prime ideal of A, and let x be a nilpotent element of A. We claim that $x \in \mathfrak{p}$. Indeed, let n be the least positive integer for which $x^n \in \mathfrak{p}$. (Such an integer exists, since x is nilpotent.) Suppose that $n > 1$. Then $x \cdot x^{n-1} \in \mathfrak{p}$; since $\mathfrak{p}$ is a prime ideal, then $x \in \mathfrak{p}$ or $x^{n-1} \in \mathfrak{p}$, a contradiction. This proves that

$$\mathfrak{N} \subset \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p}.$$

To show the converse, it suffices to show that for any non-nilpotent element a, there is some prime ideal that does not contain a.

So suppose that a is an element of A that is not nilpotent. Let S be the set of ideals of A that do not contain any element of the form a^n. Since $(0) \in S$, S is not empty; then by Zorn's Lemma, S has a maximal element $\mathfrak{m}$.

It suffices to show that $\mathfrak{m}$ is a prime ideal. Indeed, suppose otherwise; then there exist elements $x, y \notin \mathfrak{m}$ for which $xy \in \mathfrak{m}$.

$\mathfrak{m} \subsetneq \mathfrak{m} + xA, \mathfrak{m} \subsetneq \mathfrak{m} + yA$ but $a^m \in \mathfrak{m} + xA, a^n \in \mathfrak{m} + yA$ for some $m, n \in \mathbb{N}$ as $\mathfrak{m} + xA, \mathfrak{m} + yA \notin S$. Then $a^{m+n} = a^m \cdot a^n = (\mathfrak{m} + xa) \cdot (\mathfrak{m} + yb)$ for $a, b \in R$ and hence $a^{m+n} \in \mathfrak{m}$ follows from the fact that $xy \in \mathfrak{m}$ a contradiction.

□

Definition 5.13. *The* **Jacobson radical J** *of a commutative ring R with unity is defined to be the intersection of all maximal ideals of R.*

What is the Jacobson radical of $\mathbb{Z}$.
The Jacobson radical of $\mathbb{Z}/12\mathbb{Z}$ is $6\mathbb{Z}/12\mathbb{Z}$ which is the intersection of $2\mathbb{Z}/12\mathbb{Z}$ and $3\mathbb{Z}/12\mathbb{Z}$.

Proposition 5.14. *Let $\mathbf{J}$ be the Jacobson radical of a commutative ring R with unity. Then $x \in \mathbf{J}$ if and only if $1 - xy$ is unit for all $y \in R$.*

Proof. Suppose $x \in \mathbf{J}$ and $1 - xy$ is not a unit. Then $1 - xy \in M$ for some maximal ideal M. Then $1 \in M$ as $x \in \mathbf{J} \subseteq M$. Conversely suppose $x \notin M$ for some maximal ideal M then $M + \langle x \rangle = R$. Then $1 = m + xy$ form $m \in M$ and $y \in R$. Or equivalently $1 - xy \in M$. Thus $1 - xy$ is not unit in R. □

That is $\mathbf{J} = \{x \in R | 1 - xy \text{ is unit for all } y \in R\}$.

Corollary 5.15. *Let M be a maximal ideal of R and $1 + x$ is unit for every $x \in M$. Then $M = \mathbf{J}$. Further show that M is unique maximal ideal in R*

Proof. By definition $\mathbf{J} \subseteq M$. Let $x \in M$. Then $xy \in M$ for every $y \in R$. But then $1 + xy$ is a unit. That is equivalent to $x \in \mathbf{J}$.

Now we claim that $R \setminus M = U(R)$. Suppose $a \in R \setminus M$, then $\langle a \rangle + M = R$. Then $ay = 1 - m$ for some $m \in M$ and $y \in R$. But from hypothesis $1 + (-m)$ is a unit. □

The radical of an ideal plays an important role in commutative algebra, when we are concerned with the geometry aspects. This is due to the bijection existing between varieties and radical ideals. In particular, radicals are an important part of the statement of Hilbert's Nullstellensatz.

Recall radical ideal of I denoted $\sqrt{I}$ or $r(I)$ and defined

$$\sqrt{I} = \{x \in R | x^n \in I, \text{for some } n \in \mathbb{N}\}.$$

$$\begin{aligned}
\sqrt{I} &= \{x \in R | x^n \in I, \text{for some } n \in \mathbb{N}\} \\
&= \{x \in R | x^n + I = I, \text{for some } n \in \mathbb{N}\} \\
&= \{x \in R | (x + I)^n = I, \text{for some } n \in \mathbb{N}\} \\
&= \{x \in R | \ x + I \text{ is nilpotent element in } R/I\}.
\end{aligned}$$

Theorem 5.16. *1. Let R be a commutative ring and I be an ideal of R. Then $\sqrt{I}$ is an ideal of R.*

2. Let I, J be two ideals of R.

(a) If $I \subseteq J$, then $\sqrt{I} \subseteq \sqrt{J}$.

(b) $\sqrt{I} = \sqrt{\sqrt{I}}$.

(c) $\sqrt{I + J} = \sqrt{\sqrt{I} + \sqrt{J}}$

(d) $\sqrt{I \cap J} = \sqrt{I} \cap \sqrt{J}$.

3. *Let R be a commutative ring. Then I is radical ideal ($I = \sqrt{I}$) of R if and only if the only nilpotent element in R/I is I.*

4. *Let R be a commutative ring. Then every prime ideal is radical ideal.*

5. *Let R and S be ring and $f : R \to S$ be a surjective homomorphism.*

   (a) *If I is an ideal of R with $ker(f) \subseteq I$, then $f(\sqrt{I}) = \sqrt{(}f(I))$.*

   (b) *If J is an ideal of S, then $f^{-1}(\sqrt{J}) = \sqrt{(}f^{-1}(J))$.*

Proof. Proof of Part 1: Since $0 \in I \subseteq \sqrt{(}I) = r(I)$ hence $r(I) \neq \emptyset$. Let $x, y \in r(I)$, then there exists $m, n \in \mathbb{N}$ such that $x^m, y^n \in I$ now it is easy to see $(x-y)^{m+n} \in I$, $(rx)^m = r^m x^m \in I$ for every $r \in R$.
Proof of Part 2a: Let $x \in r(I)$, then there exists $m \in \mathbb{N}$ such that $x^m \in I \subseteq J$. Hence $x \in r(J)$.
Proof of Part 2b: Since $I \subseteq r(I)$ from Part 2a $r(I) \subseteq r(r(I))$. Suppose $x \in r(r(I))$ then then there exists $m, n \in \mathbb{N}$ such that $x^m \in r(I)$ and $(x^m)^n = x^{m+n} \in I$. that is $x \in r(I)$.
Proof of Part 2c: Since $I \subset r(I)$ and $J \subseteq r(J)$ hence $I + J \subseteq r(I) + r(J)$ from Part 2a $r(I+J) \subseteq r(r(I) + r(J))$. To prove other direction observe $I, J \subseteq I+J$ hence $r(I), r(J) \subseteq r(I+J)$. Then $r(I)+r(J) \subseteq r(I+J)$ again from Part 2a $r(r(I) + r(J)) \subseteq r(r(I+J)) = r(I+J)$ form Part 2b.
Proof of Part 2d: Let $a \in r(I \cap J)$, then there exists $n \in \mathbb{N}$ such that $a^n \in I \cap J$ then $a \in r(I), r(J)$ thus $r(I \cap J) \subseteq r(I) \cap r(J)$. Now suppose $b \in r(I) \cap r(J)$, then there exists $m, n \in \mathbb{N}$ such that $b^m \in I, b^n \in J$ then $b^{m+n} \in I \cap J$ which is same as $b \in r(I \cap J)$.
Proof of Part 3: Suppose $I = r(I)$ and $a + I \in R/I$ is nilpotent. Then $a^n \in I$ for some n. Thus $a \in I$. Hence only nilpotent element in R/I is I. Conversely suppose I is the only nilpotent element in R/I. We required to prove $I = r(I)$. Sufficient to prove $r(I) \subseteq I$. Let $a \in r(I)$, then $a + I$ is nilpotent in R/I thus $a + I = I$.
Proof of Part 4: Let I be a prime ideal of R. We required to prove $r(I) \subseteq I$. Let $x \in r(I)$. Then there exists $n \in \mathbb{N}$ such that $x^n \in I$ since I is prime ideal either $x \in I$ or $x^{n-1} \in I$ by induction we can prove $x \in I$.
Another way to prove is I prime ideal, then R/I is integral domain, hence only nilpotent element in R/I is I. Hence result follows from Part 3.
Proof of Part 5a:
Proof of Part 5b: □

Recall if I, J are two ideals of a commutative ring R with unity. Then the ideal quotient of colon ideal denoted $I : J$ and is defined as

$$I : J = \{x \in R | xJ \subseteq I\}.$$

1. If $J \subseteq I$, then $I : J = R$.

2. $10\mathbb{Z} : 2\mathbb{Z} = 5\mathbb{Z}$.

3. If $R = \mathbb{R}[x]$ and $I = \langle x^2 \rangle$ and $J = \langle x \rangle$ then $I : J = J$.

4. If $R = \mathbb{R}[x, y]$ and $I = \langle x^2 y, xy^2 \rangle$ and $J = \langle y^2 \rangle$ then $I : J = \langle x, ? \rangle$.

Proposition 5.17. *Let R be a commutative ring with unity. And I, J are ideals of R. Then $I : J$ is an ideal.*

Proof. Since $0 \in I : J$, $I : J \neq \emptyset$. Suppose $a, b \in I : J$ and $r \in R$. Then $aJ \subseteq I, bJ \subseteq I$. If $x \in (a-b)J$ then $x = (a-b)y = ay - by \in I$. $ra \in aJ \subseteq I$. $\square$

Let R and S be commutative rings with unity and let $f : A \to B$ be a homomorphism. Then

- If I is an ideal of A then $f(I)$ need not be an ideal of S. ($f : \mathbb{Z} \to \mathbb{Q}$ and $I = 5\mathbb{Z}$). But we define I^e smallest ideal of S containing $f(I)$ called extension of I in S. That is $I^e = \{\sum_{finite} s_i f(r_i) | s_i \in S, r_i \in I\}$.
- If J is an ideal of S, then $f^{-1}(J)$ is an ideal of R. Called contraction of J denoted J^c. It is easy to see that J^c is an ideal of R.
- If R is a subring of S and $f : R \to S$ is injective homomorphism. Then $I^e = IS$ and $J^c = f^{-1}(J) \cap R$.

5.2 Problems

1. Let $\phi : R \to S$ be a ring homomorphism and ϕ is onto. Then if P is a prime ideal in R, then $\phi(P)$ is a prime ideal in S.
2. $f : \mathbb{Z}_{20} \to \mathbb{Z}_4$ $f(x) = x \pmod 4$. Easy to check that f is a homomorphism. $f(5)$ is not zero divisor in $\mathbb{Z}_4$ but is a zero divisor in $\mathbb{Z}_{20}$. $g : \mathbb{Z} \to \mathbb{Z}_6, g(x) = x \pmod 6$ is homomorphism and 2 not a zero divisor in $\mathbb{Z}$ but $2 \pmod 6$ is a zero divisor in $\mathbb{Z}_6$.
3. Let R be a commutative ring with unity and every proper ideal is a prime ideal. Then R is a field.

   Proof. Since $\{0\}$ is proper ideal and which is prime ideal by hypothesis, R is an integral domain. Let $0 \neq a \in R$. We need to show that a is invertible. If $\langle a^2 \rangle = R$, then there exists $b \in R$ such that $a^2 b = 1$ that is $a(ab) = 1$.
   Suppose $\langle a^2 \rangle$ is proper ideal. Then $\langle a^2 \rangle$ is prime ideal. Now $a^2 \in \langle a^2 \rangle$ hence $a \in \langle a^2 \rangle$. That is $a = a^2 b$ and result follows from the fact that R is ID. $\square$
4. Let R be a commutative ring with unity. Then the following statements are equivalent.
   - The ideal P is a prime ideal.
   - For any two ideals I, J if $IJ \subseteq P$, then either $I \subseteq P$ or $J \subseteq P$.

   Proof. Suppose P is a prime ideal and $IJ \subseteq P$. Assume that $I \nsubseteq P$ or $J \nsubseteq P$. Then there exists $a \in I \setminus P$ and $b \in J \setminus P$. Then $ab \in IJ \in P$. which is contradiction for P is prime ideal. Conversely suppose that, for any two ideals I, J if $IJ \subseteq P$, then either $I \subseteq P$ or $J \subseteq P$. Let $ab \in P$. Consider $I = \langle a \rangle, J = \langle b \rangle$. Then $IJ \subseteq P$. Hence result follows from hypothesis. $\square$
5. If $a \in R$ is irreducible, then find all its divisors. Is it finite or infinite.?
6. Let I and J are ideals in R. Then the map $f : R \to R/I \times R/J$ defined by $f(x) = (x + I, x + J)$ is a ring homomorphism with kernal $I \cap J$. Further if I and J are co maximal ($I + J = R$), then f is surjective and $I \cap J = IJ$. f is injective if and only if $I \cap J = \{0\}$.

Proof.

$$f(x+y) = ((x+y)+I, (x+y)+J) = (x+I+y+I, x+J+y+J) = (x+I, x+J)+(y+I, y+J).$$

$$f(xy) = ((xy)+I, (xy)+J) = ((x+I)(y+I), (x+J)(y+J)) = (x+I, x+J)(y+I, y+J).$$

It is easy to see that kernal of f is $I \cap J$.

Now suppose if $I+J=R$, then it is easy to see $IJ = I \cap J$.

Let $(a+I, b+J) \in R/I \times R/J$. Since $I+J=R$, there exists $x \in I, y \in J$ such that $x+y=1$. Then $f(bx+ay) = (a+I, b+J)$ as
$bx+ay = bx+a(1-x) = a+\underbrace{(b-a)x}_{\in I}$ also $bx+ay = b+\underbrace{(a-b)y}_{\in J}$. The last part follows easily as *kernal of* $f = I \cap J$. □

Note this result can be extended to any number of ideals $I_1, I_2, \ldots, I_n$, where I_r, I_s are co maximal whenever $i \neq j$. For example, if $(m,n)=1$ then $\mathbb{Z}/mn\mathbb{Z} \cong (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})$.

7. Let R and S be commutative rings with unity. Suppose $f: R \to S$ be a homomorphism and q is prime ideal of S, then show that $f^{-1}(q)$ is prime ideal in R. Show that this is not true for maximal ideals.
   ($R = \mathbb{Z}, S = \mathbb{Q}, q = \langle 0 \rangle$, then q is maximal in S but $f^{-1}(q) = \langle 0 \rangle \subset \mathbb{Z}$ is prime ideal but not maximal.)

8. Let R be a PID. Then prove that every nonzero prime ideal in R is maximal.
   *Solution:*Let $P = \langle p \rangle$ be nonzero prime ideal in R. Suppose $M = \langle m \rangle$ be an ideal of R with $P \subseteq M \subseteq R$. Since $P \subseteq M$, $p \in \langle m \rangle$ That is $p = mx$ for some $x \in R$. Now $p|m$ or $p|x$ if $p|m$, then $P = M$ we are done. If $p|x$, then $x = p\ell$ for some $\ell \in R$ then $p = mp\ell$ equivalently m is a unit. Thus $M = R$.

9. Let R be a commutative ring with unity. Then show that $R[x]$ is PID if and only if R is field.
   *Solution:*If R is a field, then we know $R[x]$ is ED hence PID. Conversely suppose $R[x]$ is PID. Since R is subset of $R[x]$ hence R is integral domain. The ideal $\langle x \rangle$ is a prime ideal in $R[x]$ and is nonzero. From above problem every nonzero prime ideal in PID is maximal. Hence $R \cong R[x]/\langle x \rangle$ is field.

10. Let R be a commutative ring with unity.

    (a) Let $I_1, I_2, \ldots, I_n$ be ideals and P be a prime ideal such that $\cap_{j=1}^n I_j \subseteq P$. Then $I_k \subseteq P$ for some k.

    (b) Let $P_1, P_2, \ldots, P_n$ be prime ideals and let I be an ideal contained in $\cup_{i=1}^n P_i$. Then $I \subseteq P_i$ for some i.

    Proof. Proof of Part 1. Suppose $I_k \nsubseteq P$ for every k. Then there exists $a_k \in I_k$ but $a_k \notin P$. Let $a = a_1 \cdot a_2 \cdots a_k$ then $a \in \cap_{j=1}^n I_j \subseteq P$ a contradiction for P is prime.
    Proof of Part 2. We will prove by induction on n. If $n=1$ clearly true. Suppose $I \subseteq \cup_{i=1}^n P_i$. If for any i $I \subseteq \cup_{j \neq i} P_j$ then result holds by induction. Assume $I \subseteq \cup_{j \neq i} P_j$ for every i, then for each i there exists $a_i \in I$ but $a_i \notin P_1 \cup P_{i-1} \cup P_{i+1} \cup \ldots P_n$. that is $a_i \notin P_j$ for $j \neq i$. It is easy to see that $a_i \in P_i$ as $I \subseteq \cup_{i=1}^n P_i$. Then the element $a = a_2 a_3 \cdots a_n + a_1 a_3 \ldots a_n + \cdots + a_1 a_2 \cdots a_{n-1} \in I$ but not in $\cup_{i=1}^n P_i$ a contradiction. □

11. Let R be set of continuous real valued function on $[0,2]$. Let $I = \{f \in R | f(1) = 0\}$ show that I is maximal ideal.

Proof. Define a mapping $\phi : R \to \mathbb{R}$ by $\phi(f) = f(1)$. Show that ϕ is homomorphism and $I = ker\ \phi$. Show ϕ is surjective by taking constant mapping. $\square$

12. Give examples of Integral domains which does not possesses any irreducible elements.

    (a) Let $\bar{\mathbb{Q}}$ denotes the set of all algebraic numbers over $\mathbb{Q}$. And $\bar{\mathbb{Z}} = \bar{\mathbb{Q}} \cap \mathbb{Z}$ is the set of all algebraic integers. Then $\bar{\mathbb{Z}}$ is an integral domain but none of whose elements are irreducible.
    Check with $\langle 2^{\frac{1}{2}} \rangle \subseteq \langle 2^{\frac{1}{4}} \rangle \subseteq \langle 2^{\frac{1}{8}} \rangle \subseteq \dots$

    (b) Let D be the domain of polynomials in positive rational powers of x over $\mathbb{C}$ that is

    $$D = \{a_1 x^{r_1} + a_2 x^{r_2} + \cdots + a_n x^{r_n} | n \in \mathbb{N}, a_1, a_2, \dots, a_n \in \mathbb{C}, r_1, r_2, \dots, r_n \in \mathbb{Q}, 0 \leq r_1 < r_2 < r_2 < \cdots < r_n\}.$$

    Then show that $U(D) = \mathbb{C} \setminus \{0\}$ and D does not possesses any irreducible elements.

Proof. Suppose that $f(x) = a_1 x^{r_1} + a_2 x^{r_2} + \cdots + a_n x^{r_n}$ is an irreducible element of D. As $f(x)$ is a nonzero element of D, we may suppose $a_n \neq 0$.

$n = 1, r_1 = 0$ Then $f(x) = a_1 \neq 0$ is unit in D a contradiction.

$n = 1, r_1 > 0$ Then $f(x) = a_1 x^{r_1} = a_1 x^{\frac{r_1}{2}} x^{\frac{r_1}{2}}$ a contradiction hence $n \geq 2$ and $r_n > 0$.

Let t be the lcm of denominators of rationals $r_1, r_2, \dots, r_n$. Then t is a positive integer and $0 \leq tr_1 < tr_2 < \cdots < tr_n$ are integers. Further $f(x^t) = a_1 x^{tr_1} + a_2 x^{tr_2} + \cdots + a_n x^{tr_n}$ hence from Fundamental theorem of algebra there exists $b_1, b_2, \dots, b_n \in \mathbb{C}$ such that $f(x^t) = a_n(x - b_1) \cdot (x - b_2) \cdots (x - b_n)$. Consequently

$$f(x) = a_n(x^{\frac{1}{t}} - b_1) \cdot (x^{\frac{1}{t}} - b_2) \cdots (x^{\frac{1}{t}} - b_n).$$

But then $x^{\frac{1}{t}} - b_1 = (x^{\frac{1}{2t}} - b_1^{\frac{1}{2}})(x^{\frac{1}{2t}} - b_1^{\frac{1}{2}})$ giving a contradiction that $f(x)$ is irreducible. $\square$

13. Example of a ring in which some element is neither prime nor a product of irreducibles.

Let $R = \mathbb{Z} + x\mathbb{Q}[x] \subseteq \mathbb{Q}[x]$ be the set of all rational polynomials whose constant term is an integer.

    (a) Prove that R is an integral domain and that the units in R are ± 1.

    (b) Show that the irreducible elements in R are $\pm p$ for prime integers p and the irreducible polynomials $p(x) \in \mathbb{Q}[x]$ whose constant coefficient is ± 1.

    (c) Show that $x \in R$ cannot be written as a product of irreducibles. and conclude that R is not a unique factorization domain.

    (d) Show that $x \in R$ is not prime and describe the quotient $R/\langle x \rangle$.

Solution:

- As a subring of the integral domain $\mathbb{Q}[x], R$ cannot contain zero divisors and thus is also an integral domain.

- Now suppose $a(x)b(x) = 1$ with $a, b \in R$. Computing the degrees of both sides, we see that a and b are constants, and thus integers. So $a, b = \pm 1$. Hence the units in R are 1 and -1.
- Let $p(x) \in R$ be irreducible.
  - If $\mathsf{deg}(p(x)) = 0$, then $p \in \mathbb{Z}$. Then whenever $ab = p$, without loss of generality $a = \pm 1$. Moreover, computing degrees, $b \in \mathbb{Z}$. Thus p is a prime integer.
  - Suppose now that $p(x)$ has positive degree. Suppose further that $p(x) \in \mathbb{Q}[x]$ is reducible; say $p = ab$ where $a, b \in \mathbb{Q}[x]$. We may also write $p(x) = qp'(x)$, where $q \in \mathbb{Q}$ and $p'(x) \in \mathbb{Z}[x]$. Then $p' = q^{-1}a(x)b(x)$ is reducible in $\mathbb{Q}[x]$. By Gauss' Lemma, $p' = a'(x)b'(x), with a', b' \in \mathbb{Z}[x]$. Then $p(x) = qa'(x)b'(x)$. In particular, the constant coefficient qa_0b_0 is an integer, where a_0 and b_0 are integers and q is rational. Suppose q is given in lowest terms and write the denominator of q as q_aq_b, where $q_a|a_0$ and $q_b|b_0$. Let q_n be the numerator of q. Then $p(x) = (\frac{q_n}{q_a}a(x))(\frac{1}{q_b}b(x))$ is a factorization of $p(x) in R$, and thus $p(x) \in R$ is reducible – a contradiction. So $p(x) \in \mathbb{Q}[x]$ must be irreducible. Now write $p(x) = a + xp'(x)$. If $|a| \neq 1$, then there exists a prime integer q dividing a; say $a = qb$. Then $p(x) = q(b + \frac{1}{q}xp'(x))$. Neither of these factors is a unit, so that $p(x)$ is not irreducible after all – a contradiction. Thus $|a| = 1$ as desired.
- Suppose $x = \prod a_i$ is a product of irreducibles a_i. Since the degree of a product is the sum of degrees, exactly one of the a_i is linear – say a_0 – and the rest are constants. By part (2), $a_0 = ax \pm 1$. But then $\prod a_i$ has a nonzero constant term, a contradiction. Thus x cannot be written as a product of irreducibles in R.
- Now we show that $x \in R$ is not prime. Note that $2 \cdot \frac{1}{2}x = x$, and both of these factors is in R. However, 2 is not divisible by x (computing degrees). Suppose $\frac{1}{2}x \in (x)$. Then $\frac{1}{2}x = ax$, and since R is an integral domain, $a = \frac{1}{2} \in R$, a contradiction. So x is not prime in R.
- We claim that $R/(x) = \{a + bx + (x) \mid a \in \mathbb{Z}, b \in \mathbb{Q}, 0 \leq b < 1\}$. The $(\supseteq)$ direction is clear. To see $(\subseteq)$, suppose $p(x) + (x) \in R/(x)$ with $p(x) = \sum a_ix^i$. All powers of x of exponent at least 2 are in (x). Moreover, if a_1x is the linear coefficient of $p(x)$, we can write $a_1x = a_1'x + kx$ where $k \in \mathbb{Z}$ and a_1' is in the interval $[0, 1)$. Since $kx \in (x)$, we have $p(x) + (x) = a_0 + a_1'x + (x)$, as desired. Next we show that these representatives give distinct cosets. Suppose $a_1 + b_1x + (x) = a_2 + b_2x + (x)$; then $(a_1 - a_2) + (b_1 - b_2)x \in (x)$. Since every element of (x) has zero constant term, $a_1 = a_2$. Now $(b_1 - b_2)x \in (x)$, so that $(b_1 - b_2)x = q(x)x$. Computing degrees, in fact q is constant and we have $b_1 - b_2 \in \mathbb{Z}$. Thus $b_1 - b_2 = 0$, so that $b_1 = b_2$.

Ring	Spectrum of R (all prime ideals of R)
Field	$\{0\}$
$\mathbb{Z}$	$\{0\}, \langle p \rangle$, where p is a prime integer.
$\mathbb{K}[x]$, where $\mathbb{K}$ is a field	$\{0\}, \langle f(x) \rangle$, where $f(x)$ is an irreducible polynomial .
$\mathbb{K}[x]$, where $\mathbb{K}$ is an algebraically closed field	$\{0\}, \langle x - a \rangle$, where $a \in \mathbb{K}$ equivalently $\{0\}, \mathbb{K}$
$\mathbb{Z}[x]$	*$\{0\}, \langle p \rangle$, where p is a prime integer. * $\langle f(x) \rangle$, where $f(x)$ is irreducible in $\mathbb{Z}[x]$ * $m = \langle p, f(x) \rangle$, where $f(x) \in \mathbb{Z}[x]$ whose reduction to modulo p is irreducible * m is maximal ideal and $\mathbb{Z}[x]/m$ is finite algebraic extension field of $\mathbb{Z}_p$.
$\mathbb{Z}[i]$	* $\{0\}$, * $\langle p \rangle$, where p prime of the form $4k+3$, * $\langle 1+i \rangle$, * $\langle \pi \rangle$ and $\langle \pi' \rangle$, where $\pi \cdot \pi' = p$ and p is prime in $\mathbb{Z}$ of the form $4k+1$.

- Let R be an integral domain, and every prime ideal in R is principal. Then R is a PID.

Proof.
- Let S be the set of non-principal ideals of R, and suppose that S is nonempty. Then define a inclusion partial order on S. WRT this partial order, let $\{Ci\}$ be a chain in S, and let $C = \cup C_i$. Then we know that C is an ideal. Now we show that C is not principal. Suppose that C is principal, so that $C = \langle c \rangle$. Then $c \in C_j$ for some j, so that $C \subseteq C_j$. Then C_j is principal, contradicting our hypothesis.
  Appeal to Zorn to get an inclusion-maximal nonprincipal ideal: As C isn't principal, the chain $\{C_i\}$ has an upper bound in S. By Zorn's Lemma, S contains a maximal element. Let M be one such element.
- Let $a, b \in R$ such that $ab \in M$ while $a, b \notin M$, which exist because $M \in S$ and thus is not prime. Since $a \notin M$, we have that $M \subsetneq (M, a), M \subsetneq (M, b)$. As M is maximal in terms of non-principality, we know that both (M, a) and (M, b) are principal. Suppose that $(M, a) = \langle \alpha \rangle, (M, b) = \langle \beta \rangle$.
- Let $N = \{r \in R | r(M, a) \subseteq M\}$. Note that $(M, a)(M, b) = (M^2, Ma, Mb, ab) \subseteq M$, so that $(M, b) \subseteq N$. And further, $(M, a)N \subseteq M$, and $(M, a)N$ is principal.
- If $x \in M$, we'll show that $x = s\alpha$ for some $s \in N$, and conclude that $M = (M, a)N$ is principal a contradiction: Let $x \in M$. Since $M \subset (M, a)$, we know that $x = s\alpha$ for some $s \in R$. Note that in fact $s(M, a) = s\langle \alpha \rangle \subseteq M$, so $s \in N$. Thus $M \subseteq (M, a)N$, and more importantly M is principal. This is a contradiction on the non-principality of M, so we are wrong to assume that S is nonempty. Thus R has no nonprincipal ideals and is a principal ideal domain.

$\square$

1. Let X be a non-empty set and $P(X)$ be the power set of X. Then which of the following statement is true?

(a) $(P(X), \cup, \cap)$ is a ring.

(b) $(P(X), \cap, \cup)$ is a ring.

(c) $(P(X), \Delta, \cap)$ is a ring, where Δ is the symmetric difference of sets.

(d) $(P(X), \cup, \Delta,)$ is a ring.

2. Let $Int(\mathbb{Z}) = \{f(x) \in \mathbb{Q}[x] | f(\mathbb{Z}) \subseteq \mathbb{Z}\}$. Is $Int(\mathbb{Z})$ a subring of $\mathbb{Q}[x]$.

   (a) Yes

   (b) No

3. For every positive integer n, there exists a commutative ring with n elements.

   (a) Yes

   (b) No

4. There exists a ring which properly contains $\mathbb{Z}$ and properly contained in $\mathbb{Q}$.

   (a) Yes

   (b) No

5. Let $M_n(\mathbb{R})$ denotes the set of all $n \times n$ matrices. Is $M_n(\mathbb{R})$ a ring with usual matrix addition and multiplication of two matrices A and B is defined as $A \cdot B = AB^T$.

   (a) Yes

   (b) No

6. Let R be a ring and $u \in R$ is a unit. Then $-u$ is a unit.

   (a) Yes

   (b) No

7. Let $R = \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_4$. Then the smallest positive integer n such that $nx = 0$, where $nx = \underbrace{x + x + \cdots + x}_{n \ times}$ is

   (a) 4

   (b) 2

   (c) 24

   (d) 12

   (e) No such positive integer exists.

8. A ring R is a *Boolean ring* if $x^2 = x$ for every $x \in R$. Then the number of elements in the smallest Boolean ring is

   (a) 4

   (b) 2

(c) 3

(d) 6

9. A ring R is a *Boolean ring* if $x^2 = x$ for every $x \in R$. Is Boolean ring a commutative ring?

   (a) Yes

   (b) No

10. Let S_n set of all $n \times n$ symmetric matrices of order n over $\mathbb{R}$. Is S_n a ring with usual matrix addition and multiplication?

   (a) Yes

   (b) No

11. Let $C_3 = \{A = \begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix} | a, b, c \in \mathbb{R}\}$. Then with respect to usual matrix addition, matrix multiplication and scalar multiplication, C_3 is a

   (a) Commutative ring

   (b) Vector space over R.

   (c) Both

12. Let R be a ring (with multiplicative identity).

   (a) The set of all nonzero elements of R forms a group with respect to multiplication

   (b) The set of all units in R forms a group with respect to multiplication

   (c) The set of all non zero divisors of in R forms a group with respect to multiplication

   (d) None

13. Which of the following is a unit in $\mathbb{Z}[\sqrt{3}]$?

   (a) $-7 + 4\sqrt{3}$

   (b) $5 + 3\sqrt{3}$

   (c) $-3 - 2\sqrt{3}$

14. Number of distinct roots of the polynomial $x^2 - 1$ over $\mathbb{Z}_8$ are

   (a) 1

   (b) 2

   (c) 3

   (d) 4

15. Let R be a field. Then the units of $R[X]$ are

   (a) ± 1

(b) $R \setminus \{0\}$

(c) $R[X] \setminus \{0\}$

(d) All polynomials over R with constant term is 0

16. Let $R = \mathbb{Z}_n$ and $R' = \mathbb{Z}_{2n}$ with odd n. Then

(a) $R^* = R'^*$

(b) $|R^*| = |R'^*|$

(c) $R^* \subsetneqq R'^*$

(d) None of the above

17. Let $2 < m < n$ be such that $m|n$. Then

(a) If $\gcd(a, n) = 1$, then $\gcd(a, m) = 1$.

(b) If $\gcd(a, m) = 1$, then $\gcd(a, n) = 1$.

(c) None of the above

18. Let p be an odd prime and for an integer $n \neq 0$, define $\nu_p(n) = a$ where $p^a|n$ but $p^{a+1} \nmid n$. Further put $\nu(0) = +\infty$. Let

$$R = \{\frac{a}{b} \in \mathbb{Q} : \nu(a) - \nu(b) \geq 0\}$$

Then under addition and multiplication,

(a) R is a Ring.

(b) R is not a Ring.

19. Let S be the set of all integers whose prime factors belong to the set $\{2, 3, 5, 7\}$ and define

$$R = \{\frac{a}{b} \in \mathbb{Q} : \gcd(a, b) = 1, \text{ and } b = \pm 1 \text{ or } b \in S\}.$$

We can see that R is ring under addition and multiplication, Then

(a) R is a field.

(b) Units in R consist of all rationals whose numerators are ± 1 or whose prime factors belong to $\{2, 3, 5, 7\}$.

(c) Units in R consist of all rationals whose numerators are ± 1 or whose prime factors does not belong to $\{2, 3, 5, 7\}$.

(d) None of the above.

20. Consider $R = \mathbb{Z}[\sqrt{d}]$ with $d < 0$ squarefree. Recall $N(a + b\sqrt{d}) = |a^2 - b^2 d|$ for $a + b\sqrt{d} \in R$. Let $\alpha \in R$ and $t \in \mathbb{N}, t > 1$.

(a) If $N(\alpha) = t$ has a solution, there are exactly two solutions.

(b) If $N(\alpha) = t$ has a solution, then there are more than two solutions.

21. Let $R_1 = \mathbb{Z}[\sqrt{2}]$ and $R_2 = \mathbb{Z}[\sqrt{8}]$. Then

(a) $R_1^* \subsetneq R_2^*$

(b) $R_2^* \subsetneq R_1^*$

(c) $R_1^* = R_2^*$

22. Let non-zero elements $a, b \in R$, a ring be such that $a + b = ab$. Then

    (a) $1 - a$ is a unit in R.

    (b) $1 + a$ is a unit in R.

    (c) Both of the above

    (d) None of the above

23. Recall that a *Subring* S of a ring R is a non-empty subset S of R which is also a ring under the same addition and multiplication operation in R.

    (a) $\mathbb{Z}$ is a subring of $\mathbb{Q}$.

    (b) $\mathbb{Z}_n$ is a subring of $\mathbb{Z}$ for a positive integer $n > 1$.

    (c) $2\mathbb{Z}$ is a subring of $\mathbb{Z}$.

24. If the number of associates of a nonzero element of field is finite, then the field is finite

    (a) True

    (b) False

25. Which of the following is not a divisor of 4 in $\mathbb{Z}[\sqrt{2}]$.

    (a) $\sqrt{2}$

    (b) $2(1 + \sqrt{2})$

    (c) $4(1 + \sqrt{2})^{10}$

    (d) None of the above

26. Number of associates of 11 in $\mathbb{Z}_{101}$ is

    (a) 2

    (b) 4

    (c) 100

27. Let R be a commutative ring with unity and $a|b$, where $a, b \in R$. Then number of c such that $b = ac$ is unique.

    (a) True

    (b) False

28. The set of units forms one equivalence class under the associate relation.

    (a) True

    (b) False

5.3 Recall concepts from Vector Spaces

Definition 5.18. *A non-empty set $\mathcal{U}$ is said to be a* **vector space** *over a field $\mathbb{F}$, if $\mathcal{U}$ is an abelian group under the binary operation '+' and for every $a \in \mathbb{F}, v \in \mathcal{U}$, one defines $a.v \in \mathcal{U}$, called* **scalar multiplication***, that satisfies the following four conditions*

1. *$a.(u+v) = a.u + a.v$ for all $a \in \mathbb{F}$ and $u, v \in \mathcal{U}$,*
2. *$(a+b).v = a.v + b.v$ for all $a, b \in \mathbb{F}$ and $v \in \mathcal{U}$,*
3. *$a.(b.v) = (ab).v$ for all $a, b \in \mathbb{F}$ and $v \in \mathcal{U}$, also*
4. *$1.v = v$ for all $v \in \mathcal{U}$, where 1 is the unity of $\mathbb{F}$.*

- It is important to note that the two additions, namely, addition between elements of $\mathbb{F}$ and addition between elements of $\mathcal{U}$ need not be the same.
- The binary operations '+' and '.' are called the **vector addition** and **scalar multiplication,** respectively.
- The element $a.v$ is often written as av.
- The identity element of the group $(\mathcal{U}, +)$ is called the **zero vector** of $\mathcal{U}$ and is often denoted by $\mathbf{0}$.

Definition 5.19. *A non-empty subset $\mathcal{W}$ of $\mathcal{U}$ is said to be a* **subspace** *of $\mathcal{U}$ if $\mathcal{W}$ in itself forms a vector space, under the operations of $\mathcal{U}$.*

Problem 5.20. 1. *Show that a non-empty subset $\mathcal{W}$ of $\mathcal{U}$ is a subspace of $\mathcal{U}$ if and only if $aw_1 + bw_2 \in \mathcal{W}$, for all $a, b \in \mathbb{F}$ and $w_1, w_2 \in \mathcal{W}$.*

Definition 5.21. *Let $\mathcal{U}$ and $\mathcal{V}$ be two vector spaces over a field $\mathbb{F}$. Then a mapping $T : \mathcal{U} \to \mathcal{V}$ is said to be* **linear** *if $T(au_1 + bu_2) = aT(u_1) + bT(u_2)$, for all $a, b \in \mathbb{F}$ and $u_1, u_2 \in \mathcal{U}$. If T is also a bijection, then it is called an* **isomorphism** *(of vector spaces) and in such a case, we say that $\mathcal{U}$ and $\mathcal{V}$ are isomorphic, written $\mathcal{U} \cong \mathcal{V}$.*

Definition 5.22.

1. *Let $\mathcal{U}$ be a vector space over $\mathbb{F}$ and let $v_1, \ldots, v_n \in \mathcal{U}$. Then any element of the form $a_1v_1 + \cdots + a_nv_n$, where $a_i \in \mathbb{F}$, for $i = 1, 2, \ldots, n$, is called a* **linear combination** *of $v_1, \ldots, v_n$ over $\mathbb{F}$.*
2. *Let S be a subset of $\mathcal{U}$. Then $L(S)$, called the* **linear span** *of S, is the set of all linear combinations of finite subsets of elements of S. It is easy to see that $L(S)$ forms a subspace of $\mathcal{U}$ for any subset S of $\mathcal{U}$.*
3. *A subset $\{v_1, \ldots, v_n\}$ of elements of $\mathcal{U}$ is called* **linearly independent** *if the unique solution to the linear system $a_1v_1 + \ldots + a_nv_n = \mathbf{0}$ in the unknowns $a_1, \ldots, a_n \in \mathbb{F}$ is given by $a_i = 0$, for all $i = 1, 2, \ldots, n$. Otherwise $\{v_1, \ldots, v_n\}$ is called* **linearly dependent** *set.*
4. *A vector space $\mathcal{U}$ is said to be* finite dimensional *if there exists a finite subset S of $\mathcal{U}$ such that $L(S) = \mathcal{U}$.*
5. *A subset S of $\mathcal{U}$ is called a* **basis** *of $\mathcal{U}$ if S is linearly independent and $L(S) = \mathcal{U}$.*

6. *It is well known that the cardinality of any basis of a finite dimensional vector space $\mathcal{U}$ is always the same and is called the* **dimension** *of $\mathcal{U}$, denoted $\dim(\mathcal{U})$.*

7. *Also, two finite dimensional vector spaces $\mathcal{U}$ and $\mathcal{V}$ over the same field $\mathbb{F}$ are isomorphic if and only if $\dim(\mathcal{U}) = \dim(\mathcal{V})$.*

8. *Let $\mathcal{U}$ be a vector space over $\mathbb{F}$ and let $\mathcal{U}_1, \ldots, \mathcal{U}_k$ be subspaces of $\mathcal{U}$. Then $\mathcal{U}$ is said to be the* **direct sum** *of $\mathcal{U}_1, \ldots, \mathcal{U}_k$, if every element $v \in \mathcal{U}$ can be uniquely expressed as $v = u_1 + \cdots + u_k$, for some $u_i \in \mathcal{U}_i$, $1 \leq i \leq k$. In this case, we write $\mathcal{U} = \mathcal{U}_1 \oplus \ldots \oplus \mathcal{U}_k$ and an element $v \in \mathcal{U}$ is written as $v = u_1 \oplus \ldots \oplus u_k$. If $\mathcal{U}$ is finite dimensional, then it is well known that $\dim(\mathcal{U}) = \dim(\mathcal{U}_1) + \cdots + \dim(\mathcal{U}_k)$.*

Problem 5.23. 1. *In the following problems one has to find (if exists) vector addition and scalar multiplication.*

   (a) *Is $\mathbb{Z}$ a vector space over $\mathbb{Q}$.*

   (b) *Is $\mathbb{Q}$ a vector space over $\mathbb{Z}$.*

   (c) *Is $\mathbb{R}$ a vector space over $\mathbb{Q}$?. If so what is its dimension?*

   (d) *Is $\mathbb{C}$ a vector space over $\mathbb{R}$. If so what is its dimension.*

   (e) *Let $\mathbb{F}$ be a filed. Then show that $\mathbb{F}$ is vector space over $\mathbb{F}$. In general, show that $\mathbb{K}$ is vector space over $\mathbb{F}$, whenever $\mathbb{F}$ is subfield of $\mathbb{K}$.*

   (f) *Let $\mathbb{F}$ be a field. Then show that $\mathbb{F}[x]$ is a vector space over $\mathbb{F}$ and find its dimension.*

   (g) *Show that $M_n(\mathbb{R})$ is a vector space over $\mathbb{R}$.*

2. *Show that any subset of linearly independent set is linearly independent.*

3. *Show that any super set of linearly dependent set is linearly dependent.*

4. *Let $\mathcal{U}$ be a finite dimensional vector space and S be a linearly independent subset of $\mathcal{U}$. Then show that S can be extended to form a basis of $\mathcal{U}$.*

5. *Give an example of vector space with finite number of elements.*

6. *Let X be any non-empty set. Let*
$$R_X^{\mathbb{R}} = \{f | f : X \to \mathbb{R}\}.$$
That is R_X^A be the collection of all functions from X to $\mathbb{R}$, where addition and multiplication defined component wise i.e.,
$$(f+g)(x) = f(x) + g(x) \text{ and } (fg)(x) = f(x)g(x).$$
Then Show that $R_X^{\mathbb{R}}$ is both vector space and a ring. Further show that $R_{[0,1]}^{\mathbb{R},cont}$ and $R_{[0,1]}^{\mathbb{R},diff}$ are subspaces of $R_{[0,1]}^{\mathbb{R}}$.

1. You are aware $\mathbb{Z}/p\mathbb{Z}$ is a field of characteristic p. There are only two groups up to isomorphism of order p^2. Let G be an additive group of order p^2. Let $x \in G$. Then $ord(x) \in \{1, p, p^2\}$. If there is an element $g \in G$ of order p^2. Then $G \cong \mathbb{Z}/p^2\mathbb{Z}$ with map $g^a \to a$. In this case G is not a vector space over $\mathbb{Z}/p\mathbb{Z}$ as in $\mathbb{Z}/p^2\mathbb{Z}$, there is an element x such that $px \neq 0$.

If there are no element of order p^2. Then every element except identity is of order p. In this case G is vector space over $\mathbb{Z}/p\mathbb{Z}$ isomorphism is given by $g_1^a g_2^b \to (a, b)$. Hence $G \cong (\mathbb{Z}/p\mathbb{Z})^2$. In general $(\mathbb{Z}/p\mathbb{Z})^n$ is a vector space over $\mathbb{Z}/p\mathbb{Z}$ called elementary abelian p-group.

2. $\mathbb{Z}/p\mathbb{Z}$ is a vector space over itself.

3. Every field of characteristic p contains a field isomorphic to $\mathbb{Z}/p\mathbb{Z}$. Hence the fields $\mathbb{Q}$ and $\mathbb{Z}/p\mathbb{Z}$ are called prime fields, every field of characteristic zero contains an isomorphic copy of $\mathbb{Q}$.

6 Lecture-6 (Field Extensions)

A field $\mathbb{K}$ is said to be an **extension** of a field $\mathbb{F}$, if $\mathbb{K}$ contains $\mathbb{F}$ or $\mathbb{F}$ is a subfield of $\mathbb{K}$.

- Let $\mathbb{K}$ be an extension of $\mathbb{F}$. Then $\mathbb{K}$ is a vector space over $\mathbb{F}$.
- The dimension of $\mathbb{K}$ over $\mathbb{F}$ as a vector space is denoted by $[\mathbb{K} : \mathbb{F}]$ and is called the degree of $\mathbb{K}$ over $\mathbb{F}$.
- If $[\mathbb{K} : \mathbb{F}]$ is finite, then $\mathbb{K}$ is called a **finite extension** of $\mathbb{F}$ otherwise, it is called **infinite extension**.

Problem 6.1.

1. *If $\mathbb{L}$ is a finite extension of $\mathbb{K}$ and if $\mathbb{K}$ is a finite extension of $\mathbb{F}$, then $\mathbb{L}$ is a finite extension of $\mathbb{F}$. Moreover $[\mathbb{L} : \mathbb{F}] = [\mathbb{L} : \mathbb{K}][\mathbb{K} : \mathbb{F}]$.*
2. *If $\mathbb{L}$ is a finite extension of $\mathbb{F}$ and $\mathbb{K}$ is subfield of $\mathbb{L}$ containing $\mathbb{F}$, then $[\mathbb{K} : \mathbb{F}]$ divides $[\mathbb{L} : \mathbb{F}]$.*
3. *Let $f(x) \in \mathbb{F}[x]$ be an irreducible polynomial. Then $\mathbb{F}[x]/\langle f(x)\rangle$ is a field and is a finite extension over $\mathbb{F}$ (up to isomorphism).*

Definition 6.2. *Let $\mathbb{K}$ be an extension field over filed $\mathbb{F}$. Then an element $a \in \mathbb{K}$ is said to be* **algebraic over** $\mathbb{F}$ *if there exists elements $\alpha_0, \alpha_1, \ldots, \alpha_n \in \mathbb{F}$ not all zero such that*

$$\alpha_0 + \alpha_1 a + \cdots + \alpha_n a^n = 0.$$

Problem 6.3.

1. *An element $a \in \mathbb{K}$ is algebraic over $\mathbb{F}$ if and only if there exists a positive integer n such that the set $\{1, a, a^2, \ldots, a^n\}$ is linearly dependent over $\mathbb{F}$.*
2. *An element $a \in \mathbb{K}$ is algebraic over $\mathbb{F}$ if and only if there exists a non-zero polynomial $q(x) \in \mathbb{F}[x]$ such that $q(a) = 0$.*

Let $\mathbb{K}$ be an extension of $\mathbb{F}$ and let $a \in \mathbb{K}$. Let $\mathcal{M}$ be the collection of all subfields of $\mathbb{K}$ which contains both $\mathbb{F}$ and a. Then it is clear that $\mathcal{M}$ is non-empty as it contains $\mathbb{K}$. The intersection all those subfields is the smallest field containing both $\mathbb{F}$ and a, denoted by $\mathbb{F}(a)$.

Definition 6.4.

- *An extension field $\mathbb{K}$ of $\mathbb{F}$ is said to be an* **algebraic extension** *if every element of $\mathbb{K}$ is algebraic over $\mathbb{F}$, i.e., if every element of $\mathbb{K}$ is a zero of some non-zero polynomial with coefficients in $\mathbb{F}$.*
- *The monic polynomial $q(x) \in \mathbb{F}[x]$ of least degree satisfying $q(\alpha) = 0$ is called the* **minimal polynomial** *of α over $\mathbb{F}$.*
- *The element $a \in \mathbb{K}$ is said to be* **algebraic of degree** *n over $\mathbb{F}$ if the degree of its minimal polynomial is n.*
- *A complex number is said to be an* **algebraic number** *if it is algebraic over the field of rational numbers.*

- *An algebraic number a is said to be an* **algebraic integer** *if it satisfies an equation of the form*

$$\alpha_0 + \alpha_1 a + \alpha_2 a^2 + \cdots + \alpha_{n-1} a^{n-1} + a^n = 0,$$

where $\alpha_0, \alpha_1, \ldots, \alpha_{n-1} \in \mathbb{Z}$.

Problem 6.5. 1. *The element $a \in \mathbb{K}$ is algebraic over $\mathbb{F}$ if and only if $\mathbb{F}[a]$ is a finite extension of $\mathbb{F}$.*

2. *Let $\alpha \in \mathbb{K}$ be algebraic over $\mathbb{F}$ of degree n. Then*
   (a) $\mathbb{F}(\alpha) = \left\{ \frac{f(\alpha)}{g(\alpha)} | f(x), g(x) \in \mathbb{F}[x], g(\alpha) \neq 0 \right\}$.
   (b) $\mathbb{F}(\alpha) = \{a_0 + a_1\alpha + a_2\alpha^2 + \cdots + a_{n-1}\alpha^{n-1} | a_0, a_1, \ldots, a_{n-1} \in \mathbb{F}\}$

3. *Let $q(x)$ be the minimal polynomial of $a \in \mathbb{K}$ over $\mathbb{F}$. Then show that $q(x)$ is irreducible. Further show that if $f(a) = 0$ for some $f(x) \in \mathbb{F}[x]$, then $q(x)$ divides $f(x)$.*

4. *Show that every finite extension is also algebraic extension. what about the converse?*

5. *Let $\mathbb{F}$ be a subfield of $\mathbb{C}$ and let $q(x) \in \mathbb{F}[x]$ be the minimal polynomial of $\alpha \in \mathbb{C}$. Then the field $\mathbb{F}(\alpha)$ coincides with the ring $\mathbb{F}[\alpha]$ and $\mathbb{F}[x]/\langle q(x) \rangle \cong \mathbb{F}[\alpha]$.*

6. *If $a \in \mathbb{K}$ is algebraic of degree n over $\mathbb{F}$, then $[\mathbb{F}[a] : \mathbb{F}] = n$.*

7. *If $a, b \in \mathbb{K}$ are algebraic over $\mathbb{F}$ then $a \pm b, ab$ and a/b (if $b \neq 0$) are all algebraic over $\mathbb{F}$. In other words, the elements in $\mathbb{K}$ which are algebraic over $\mathbb{F}$ form a subfield of $\mathbb{K}$.*

8. *If $a, b \in \mathbb{K}$ are algebraic over $\mathbb{F}$ of degree m and n, respectively, then $a \pm b, ab, a/b$ (if $b \neq 0$) are algebraic over $\mathbb{F}$ of degree at most mn.*

9. *If $\mathbb{L}$ is a algebraic extension of $\mathbb{K}$ and if $\mathbb{K}$ is a algebraic extension of $\mathbb{F}$, then $\mathbb{L}$ is a algebraic extension of $\mathbb{F}$.*

10. *Show that $\sqrt{2}, \sqrt{3}, \sqrt{2}\sqrt{3}, \sqrt{2} + \sqrt{3}, \sqrt{2}/\sqrt{3}$ are algebraic over $\mathbb{Q}$.*

11. *If a rational number r is also an algebraic integer, then show that $r \in \mathbb{Z}$.*

12. *Prove that sum and product of two algebraic integers is again a algebraic integer.*

13. *Show that $\mathbb{Q} \subset \mathbb{Q}[\sqrt{2}] \subset \mathbb{Q}[\sqrt[6]{2}]$.*

14. *Show that $\mathbb{Q}[\sqrt[n]{2}] : \mathbb{Q}] = n$.*

15. *Show that the map $a + b\sqrt{2} \to a - b\sqrt{2}$ is an isomorphism in $\mathbb{Q}[\sqrt{2}]$.*

16. *Prove that $\mathbb{Q}[\sqrt{2} + \sqrt{3}] = \mathbb{Q}[\sqrt{2}, \sqrt{3}]$, where $\mathbb{F}[a, b] = \mathbb{F}[a][b]$.*

17. *Find $[\mathbb{Q}[\sqrt{3 + 2\sqrt{2}}] : \mathbb{Q}]$.*

18. *Find the minimal polynomial of $1 + i$ over $\mathbb{Q}$ and over $\mathbb{C}$.*

19. *If $[\mathbb{F}[\alpha] : \mathbb{F}]$ is odd, then show that $\mathbb{F}[\alpha] = \mathbb{F}[\alpha^2]$.*

20. *If $\mathbb{K}$ is an algebraic extension over $\mathbb{F}$ and R be a ring such that $\mathbb{F} \subseteq R \subseteq \mathbb{K}$, then R is a field.*

21. *If $p \in \mathbb{Z}^+$ is a prime, then $x^3 - p$ is irreducible over $\mathbb{Q}[i]$.*

22. *If $\mathbb{K} = \mathbb{Q}[\alpha_1, \alpha_2, \ldots, \alpha_n]$, where $\alpha_i^2 \in \mathbb{Q}$, $1 \leq i \leq n$, then $\sqrt[3]{2} \notin \mathbb{K}$.*

Problems with solutions

1. Let R be an ID. Then R is also ED if
   *for every every $a \neq 0$ in R there is defined a non-negative integer $d(a)$ such that
   (a) for all $a, b \in R$, both non-zero, $d(a) \leq d(ab)$.
   (b) for any $a, b \in R$, both non-zero, there exists $q, r \in R$ such that $a = bq + r$, where either $r = 0$ or $d(a) < d(b)$.

2. Give an example of a polynomial in $\mathbb{Z}[x]$ which is reducible in $\mathbb{R}[x]$ but irreducible in $\mathbb{Q}[x]$
   * $x^2 - 2$.

3. Give an example of a polynomial in $\mathbb{Z}[x]$ which is reducible in $\mathbb{Z}[x]$ but irreducible in $\mathbb{Q}[x]$
   *$2x + 2$. As $2x + 2 = 2(x + 1)$ is a product of two polynomials but none of them are units in $\mathbb{Z}$ (2 does not have inverse in $\mathbb{Z}$).

4. Give an example of ED which is not field.
   *$\mathbb{Z}$ or $\mathbb{F}[x]$.

5. Give an example of PID which is not ED.

   * $\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$. I know, now you are thinking sir (me) not told this in the class but asked in the exam. But recall, from the day one, this course started I am informing you that in this course we see $ID \subseteq UFD \subseteq PID \subseteq ED \subseteq \ field$.

6. Which of following two polynomials are irreducible over $\mathbb{Q}$

$$a) x^3 - 2x^2 - 2x \qquad b) x^{100} - 3x^{50} - 3x^{25} - 3x^{10} - 3x^3 - 3x^2$$

   .

   - None of them.
   - As $x^3 - 2x^2 - 2x = x(x^2 - 2x - 2)$ which is product two polynomials none of them is a unit.
   - Similarly second one.
   - Those who applied Eisenstein criteria have to think what is a_0 here and has to recall every number divides zero.

7. Give an example of ID which is not ED [also not a PID (otherwise you can give same example as in Q5)]
   *$\mathbb{Z}[x]$ (Any body who observes carefully can get this answer from next question.)

8. Give an example for ring $R \neq \mathbb{Z}$ such that R is PID but $R[x]$ is not.

   - Many times we discussed in the class that $\mathbb{Z}$ and $\mathbb{F}[x]$ behaves almost in same manner.

- We know that Z is PID but $\mathbb{Z}[x]$ (is UFD) not a PID.
- That is $\mathbb{F}[x, y] = \mathbb{F}[y][x]$ is ID but not PID, hence not a ED.

9. Provide an irreducible polynomial over $\mathbb{F}$, where $\mathbb{F}$ is any field.

   - x.
   - Which is a polynomial of degree 1, hence irreducible, as one can't write it as product to two lower degree polynomials.
   - Its only root is 0, which belongs to every field.
   - One can also choose $x - a$ for $a \in \mathbb{F}$.
   - Every field contains 0 and 1 so x and $x - 1$ are irreducible over any field $\mathbb{F}$.
   - In case of $\mathbb{Z}_2$ these are the only irreducible polynomials.

10. Let $a \in \mathbb{F}$ be a non-zero element. Then find all its associates.
    * Every non-zero element of $\mathbb{F}$ is an associate of a.

11. State the Fundamental Theorem of Algebra in terms of irreducible polynomials over $\mathbb{C}$.
    *The only irreducible polynomials in $\mathbb{C}$ are of degree 1.

12. Why $a \in \mathbb{F}$ is algebraic over $\mathbb{F}$?.

    - Because it satisfies the polynomial $x - a$ which belongs to $\mathbb{F}[x]$ and non-zero.

13. Give an example of a polynomial of degree n in $\mathbb{Q}[x]$ which is irreducible over $\mathbb{Q}$.
    *$x^n - 2$

14. Is it possible to find polynomial of degree 3 in $\mathbb{R}[x]$ which is also irreducible over $\mathbb{R}$. Explain.

    - No.
    - We know that $\mathbb{C}$ is a vector space over $\mathbb{R}$ with basis $\{1, i\}$. And every non-zero non-constant polynomial over $\mathbb{R}$ has a root in $\mathbb{C}$ (Fundamental theorem of algebra).

15. Find $[Q[\sqrt{2} + \sqrt{5}] : \mathbb{Q}]$ (ans:4).

16. Give an example of a vector space ($\neq \{0\}$) with finite number of elements.
    $\mathbb{Z}_p$ is a vector space over $\mathbb{Z}_p$ with p elements.

17. State Eisenstein criteria.
    Let $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ be a polynomial with integer coefficients. Suppose that for some prime number p,

    - $p \nmid a_n$
    - $p|a_1, p|a_2, \ldots, p|a_0$
    - $p^2 \nmid a_0$.

    Then $f(x)$ is irreducible over the rationals.

18. State Gauss lemma.
    The product of two primitive polynomials is again a primitive polynomial.

19. Find the minimal polynomial of $2 - i$ over $\mathbb{Q}$ and over $\mathbb{C}$.

    - The minimal polynomial of $2 - i$ over $\mathbb{Q}$ is $x^2 - 4x + 5$.
    - The minimal polynomial of $2 - i$ over $\mathbb{C}$ is $x - 2 + i$.

20. Is every finite extension is an algebraic extension? explain.

    - Yes.
    - Let $\mathbb{K}$ be a finite extension over $\mathbb{F}$ and $[\mathbb{K} : \mathbb{F}] = n$. So $\mathbb{K}$ is a vector space over $\mathbb{F}$ and any basis of $\mathbb{K}$ over $\mathbb{F}$ contains n elements. Consequently, any set with $n + 1$ or more elements in $\mathbb{K}$ is linearly dependent over $\mathbb{F}$.
    - Now we claim that $\mathbb{K}$ is an algebraic extension over $\mathbb{F}$. That means we have to show that every element in $\mathbb{K}$ is algebraic over $\mathbb{F}$. In other words, every element in $\mathbb{K}$ is a root of a non-zero polynomial of over $\mathbb{F}$.
    - Let $a \in \mathbb{K}$. Then the set $\{1, a, a^2, \ldots, a^n\}$ is LD as it contains $n + 1$ elements. So there exists $\alpha_0, \alpha_1, \ldots, \alpha_n \in \mathbb{F}$ not all zero such that

      $$\alpha_0 + \alpha_1 a + \cdots + \alpha^n a^n = 0,$$

      or a satisfies the non-zero polynomial

      $$\alpha_0 + \alpha_1 x + \cdots + \alpha^n x^n.$$

7 Lecture-7 (continuation of Field extensions)

How to get finite extensions

Now we will see the standard way to construct a field by adjoining a root of a polynomial to a smaller field.

Theorem 7.1. *Let $f(x) \in \mathbb{F}[x]$ be an irreducible polynomial. Then there exists a field $\mathbb{K}$ containing isomorphic copy of $\mathbb{F}$ in which $f(x)$ has a root.*

Proof. Let $\mathbb{K} = \mathbb{F}[x]/\langle f(x)\rangle$. Then $\mathbb{K}$ is field follows from the fact that $\mathbb{F}[x]$ is Euclidean domain and $\langle f(x)\rangle$ is maximal ideal in $\mathbb{F}[x]$. Now we will show

- $\mathbb{K}$ contains (a field isomorphic to) $\mathbb{F}$.
- $\mathbb{K}$ contains a root α of f.

1. Let $I = \langle f(x)\rangle$. Define $\bar{\mathbb{F}} = \{a + I | a \in \mathbb{F}\}$. and a mapping $\phi : \mathbb{F} \to \bar{\mathbb{F}}$ as $\phi(a) = a + I$. Then it is easy to check ϕ is an isomorphism from $\mathbb{F}$ to $\bar{\mathbb{F}}$.
2. We take $\alpha = x + I$. Let $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$. Then

$$\begin{aligned} f(x+I) &= (a_0+I) + (a_1+I)(x+I) + (a_2+I)(x+I)^2 + \cdots + (a_n+I)(x+I)^n \\ &= a_0 + a_1x + a_2x^2 + \cdots + a_nx^n + I \\ &= I. \end{aligned}$$

□

- Our proof shows that a field with required properties exists. However it is defined as a factor ring, which is not the most convenient form for calculation. As always, calculations in a factor ring is very much easier if we make a good choice of coset representatives.
- Let f be an irreducible polynomial of degree $n > 0$ over a filed $\mathbb{F}$. We claim that every coset of $\langle f(x)\rangle$ in $\mathbb{F}[x]$ has a unique representative r satisfying $r = 0$ or $\deg(r) < n$. Such r exists because of Euclidean of $\mathbb{F}[x]$.
- Let $g(x) \in \mathbb{F}[x]$, then $g(x) = f(x)q(x) + r(x)$, where $r(x) = 0$ or $\deg(r(x)) < n$.. Then g and r differ by a multiple of f so $g + \langle f(x)\rangle = r + \langle f(x)\rangle$. If r_1, r_2 are two representatives of the same coset with $r_i = 0$ or $\deg(r_i) < n$. Then $f | r_1 - r_2$ hence $r_1 = r_2$.
- Further the coset $r + \langle f(x)\rangle$ is equal to $r(\alpha)$, where α is the coset $x + \langle f(x)\rangle$.
- Every element of $\mathbb{K} = \mathbb{F}[x]/\langle f(x)\rangle$ can be uniquely expressed in the form

$$c_0 + c_1\alpha + c_2\alpha^2 + \ldots c_{n-1}\alpha^{n-1},$$

  where $c_0, c_1, \ldots, c_{n-1} \in \mathbb{F}$.

Let $\mathbb{K}$ be an extension of $\mathbb{F}$ and let $a_1, a_2, a_3, \ldots, a_n \in \mathbb{K}$. Let $\mathcal{M}$ be the collection of all subfields of $\mathbb{K}$ which contains both $\mathbb{F}$ and $a_1, a_2, a_3, \ldots, a_n$. Then it is clear that $\mathcal{M}$ is non-empty as it contains $\mathbb{K}$. The intersection all those subfields is the smallest field containing $\mathbb{F}$ and $a_1, a_2, a_3, \ldots, a_n$, denoted by $\mathbb{F}(a_1, a_2, a_3, \ldots, a_n)$. The field $\mathbb{F}(a_1, a_2, a_3, \ldots, a_n)$ is called **finitely generated** extension of $\mathbb{F}$. If $n = 1$, then it is called **simple extension** denoted $\mathbb{F}(a)$.

- It is clear that $\mathbb{F}(a)$ is an extension over $\mathbb{F}$.
- Depending on $[\mathbb{F}(a) : \mathbb{F}]$ is finite or not we define $a \in \mathbb{K}$ as **algebraic** or **transcendental** over $\mathbb{F}$ respectively.
- For example, if $a \in \mathbb{F}$, then $\mathbb{F}(a) = \mathbb{F}$ and clearly $\mathbb{F}(a)$ is a finite extension over $\mathbb{F}$.

Closer look at simple extension

Let $\mathbb{K}$ be an extension of $\mathbb{F}$ and $a \in \mathbb{K}$. Then we define a mapping

$$\psi : \mathbb{F}[x] \to \mathbb{F}[a] \;\; by \;\; \psi(f(x)) = f(a).$$

One can easily check that ψ a ring on-to homomorphism.

$\ker \psi$ **is one-one:** That is there is no nonzero polynomial in $\mathbb{F}[x]$ satisfied by a. In this case $\mathbb{F}[x] \cong \mathbb{F}[a]$ and a is transcendental over $\mathbb{F}$. The field of fraction $F(a)$ of $\mathbb{F}[a]$ is called **purely transcendental** extension of $\mathbb{F}$. Further $[\mathbb{F}(a) : \mathbb{F}]$ is infinite. Let a and b be two transcendental elements over $\mathbb{F}$, then $\mathbb{F}[a] \cong \mathbb{F}[x] \cong \mathbb{F}[b]$. In this case $\mathbb{F}[a]$ is not field as $1/a$ does not have inverse in $\mathbb{F}[a]$.

$\ker \psi$ **is not one-one:** In this case $\ker \psi$ is set of all polynomials satisfied by a. Further $\ker \psi$ is an ideal of $\mathbb{F}[x]$. Hence from fundamental theorem of homomorphism of rings $\mathbb{F}[x]/\ker \psi \cong \mathbb{F}[a]$.

If a is any root of $f(x)$. Then from 7 $\mathbb{K} = \mathbb{F}[x]/\langle f(x)\rangle = \mathbb{F}[x]/ker\psi \cong \mathbb{F}[a]$. Since $\mathbb{F}[a]$ is field it is equal to its field of fractions $\mathbb{F}(a)$. Further $[\mathbb{F}(a) : \mathbb{F}] = \deg(f(x))$.

1. show that $\mathbb{F}(\alpha) = \left\{ \frac{f(\alpha)}{g(\alpha)} | f(x), g(x) \in \mathbb{F}[x], g(\alpha) \neq 0 \right\}$.
2. If α is algebraic, then $\mathbb{F}(\alpha) = \mathbb{F}[\alpha]$.

Algebraic extension, minimal polynomial

Definition 7.2. *Let $\mathbb{K}$ be an extension field over filed $\mathbb{F}$. Then an element $a \in \mathbb{K}$ is said to be* **algebraic over** $\mathbb{F}$ *if there exists elements $\alpha_0, \alpha_1, \ldots, \alpha_n \in \mathbb{F}$ not all zero such that*

$$\alpha_0 + \alpha_1 a + \cdots + \alpha_n a^n = 0.$$

Problem 7.3.

1. *An element $a \in \mathbb{K}$ is algebraic over $\mathbb{F}$ if and only if there exists a positive integer n such that the set $\{1, a, a^2, \ldots, a^n\}$ is linearly dependent over $\mathbb{F}$.*
2. *An element $a \in \mathbb{K}$ is algebraic over $\mathbb{F}$ if and only if there exists a non-zero polynomial $q(x) \in \mathbb{F}[x]$ such that $q(a) = 0$.*
3. *Let $q(x)$ be the minimal polynomial of $a \in \mathbb{K}$ over $\mathbb{F}$. Then show that $q(x)$ is irreducible. Further show that if $f(a) = 0$ for some $f(x) \in \mathbb{F}[x]$, then $q(x)$ divides $f(x)$.*

Theorem 7.4. *Let $\alpha \in \mathbb{K}$ be algebraic over $\mathbb{F}$ of degree n. Then show that*

1. $\mathbb{F}(\alpha) = \mathbb{F}[\alpha]$, *where* $\mathbb{F}(\alpha) = \left\{ \frac{g(\alpha)}{h(\alpha)} | g(x), h(x) \in \mathbb{F}[x], h(\alpha) \neq 0 \right\}$.

   Proof. It is clear that $\mathbb{F}[\alpha] \subseteq \mathbb{F}(\alpha)$. Let $f(x)$ be the minimal polynomial of $\alpha \in \mathbb{K}$. Since $h(\alpha) \neq 0$, $\gcd(h(x), f(x)) = 1$, hence there exists $r(x), s(x) \in \mathbb{F}[x]$ such that $f(x)r(x) + h(x)s(x) = 1$. Hence $s(\alpha) = 1/h(\alpha)$. Consequently $\frac{g(\alpha)}{h(\alpha)} = g(\alpha)s(\alpha) \in \mathbb{F}[\alpha]$. □

2. $\mathbb{F}(\alpha) = \{a_0 + a_1\alpha + a_2\alpha^2 + \cdots + a_{n-1}\alpha^{n-1} | a_0, a_1, \ldots, a_{n-1} \in \mathbb{F}\}$

   Proof. Continuation to above proof $g(x)s(x) = f(x)q(x) + r(x)$ hence $g(\alpha)s(\alpha) = r(\alpha)$, where $r(\alpha) = 0$ or $\deg(r(x)) < \deg(f(x))$. Hence every element in $\mathbb{F}(\alpha)$ is of the form $a_0 + a_1\alpha + a_2\alpha^2 + \cdots + a_{n-1}\alpha^{n-1}$, where $a_0, a_1, \ldots, a_{n-1} \in \mathbb{F}$. □

Definition 7.5.

- *An extension field $\mathbb{K}$ of $\mathbb{F}$ is said to be an* **algebraic extension** *if every element of $\mathbb{K}$ is algebraic over $\mathbb{F}$, i.e., if every element of $\mathbb{K}$ is a zero of some non-zero polynomial with coefficients in $\mathbb{F}$.*

- *The monic polynomial $q(x) \in \mathbb{F}[x]$ of least degree satisfying $q(\alpha) = 0$ is called the* **minimal polynomial** *of α over $\mathbb{F}$.*

- *The element $a \in \mathbb{K}$ is said to be* **algebraic of degree** *n over $\mathbb{F}$ if the degree of its minimal polynomial is n.*

- *A complex number is said to be an* **algebraic number** *if it is algebraic over the field of rational numbers.*

- *An algebraic number a is said to be an* **algebraic integer** *if it satisfies an equation of the form*

$$\alpha_0 + \alpha_1 a + \alpha_2 a^2 + \cdots + \alpha_{n-1} a^{n-1} + a^n = 0,$$

  where $\alpha_0, \alpha_1, \ldots, \alpha_{n-1} \in \mathbb{Z}$.

Is every finite extension is an algebraic extension? explain. What about the converse?

Solution:

- Yes.

- Let $\mathbb{K}$ be a finite extension over $\mathbb{F}$ and $[\mathbb{K} : \mathbb{F}] = n$. So $\mathbb{K}$ is a vector space over $\mathbb{F}$ and any basis of $\mathbb{K}$ over $\mathbb{F}$ contains n elements. Consequently, any set with $n + 1$ or more elements in $\mathbb{K}$ are linearly dependent over $\mathbb{F}$.

- Now we claim that $\mathbb{K}$ is an algebraic extension over $\mathbb{F}$. That means we have to show that every element in $\mathbb{K}$ is algebraic over $\mathbb{F}$. In other words, every element in $\mathbb{K}$ is a root of a non-zero polynomial of over $\mathbb{F}$.

- Let $a \in \mathbb{K}$. Then the set $\{1, a, a^2, \ldots, a^n\}$ is LD as it contains $n+1$ elements. So there exists $\alpha_0, \alpha_1, \ldots, \alpha_n \in \mathbb{F}$ not all zero such that

$$\alpha_0 + \alpha_1 a + \cdots + \alpha^n a^n = 0,$$

or a satisfies the non-zero polynomial

$$\alpha_0 + \alpha_1 x + \cdots + \alpha^n x^n.$$

- Converse is not true. $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \sqrt{11}, \ldots]$ is an example of algebraic extension but not a finite extension. Another example is $\mathbb{Q}[\sqrt{2}, \sqrt[3]{2}, \sqrt[4]{2}, \sqrt[5]{2} \ldots]$. But we can't accept these as examples with out showing set of all algebraic elements in $\mathbb{K}$ over $\mathbb{F}$ forms a field.

Theorem 7.6. *The extension $\mathbb{K}/\mathbb{F}$ is finite if and only if $\mathbb{K}$ is generated by finite number of algebraic elements over $\mathbb{F}$. More precisely, a field generated over $\mathbb{F}$ by a finite number of algebraic elements of degrees $n_1, n_2, \ldots n_k$ is algebraic of degree $\leq n_1 \cdot n_2 \ldots n_k$.*

One has to show that $\mathbb{F}(a, b) = \mathbb{F}(a)(b)$. If $a_1, a_2, \ldots, a_n \in \mathbb{K}$ are algebraic over $\mathbb{F}$, then $\mathbb{F} \subseteq \mathbb{F}_1 = \mathbb{F}[a_1] \subseteq \mathbb{F}_2 = \mathbb{F}_1[a_2] \subseteq \ldots \mathbb{F}[a_n] = F_n$.

If $a, b \in \mathbb{K}$ are algebraic over $\mathbb{F}$, then $\mathbb{F}(a, b)$ is finite extension over $\mathbb{F}$. Hence $a+b, ab, a/b$ etc algebraic over $\mathbb{F}$.

Theorem 7.7. *If $\mathbb{K}$ is algebraic over $\mathbb{F}$ and $\mathbb{L}$ is algebraic over $\mathbb{K}$, then $\mathbb{L}$ is algebraic over $\mathbb{F}$.*

Compositum of two fields

Definition 7.8. *Let $\mathbb{K}_1$ and $\mathbb{K}_2$ are two subfields of a field $\mathbb{K}$. Then the composite field of K_1 and K_2, denoted K_1K_2, is the smallest field of K containing both K_1 and K_2. Similarly, the composite of any collection of subfields of K is the smallest subfield containing all subfields.*

Definition 7.9. *Any field which is a finite extension of $\mathbb{Q}$ is called* **number field**. *And is said to be* **quadratic extension** *or* **Quadratic number field** *if its degree of extension is 2.*

Theorem 7.10. *Let $\mathbb{K}$ be a quadratic extension over $\mathbb{Q}$. Then $\mathbb{K} = \mathbb{Q}[\sqrt{d}]$ where d is square free integer.*

Theorem 7.11. *Let $\overline{\mathbb{Z}}$ denotes the set of all algebraic integers. Then show that*

1. $\mathbb{Q} \cap \overline{\mathbb{Z}} = \mathbb{Z}$.
2. $\mathbb{Q}[\sqrt{d}] \cap \overline{\mathbb{Z}} = \begin{cases} \mathbb{Z}[\sqrt{d}] & \text{if } d \equiv 2 \text{ or } 3 \pmod 4 \\ \mathbb{Z}[\frac{1+\sqrt{d}}{2}] & \text{if } d \equiv 1 \pmod 4 \end{cases}$

Proof. Proof of Part-1: Already done.
Proof of Part-2: Suppose $\alpha = a + b\sqrt{d} \in O_{\mathbb{K}}$ where $a, b \in \mathbb{Q}$. Then $x^2 - 2ax + a^2 - b^2 d \in \mathbb{Z}[x]$. That is $\boxed{2a \in \mathbb{Z}}$, $4a^2 - 4b^2 d \in \mathbb{Z}$. Hence $4b^2 d \in \mathbb{Z}$. Suppose $b = \frac{r}{s}$ where $(r, s) = 1$. we claim that if $k|s$, then $k = 1$ or $k = 2$. If $s = 2^t$ then $t = 1$ as if $t > 1$ then $4|d$ which is contradiction for d is square free. Suppose p is odd prime dividing s, then again $p^2|d$ which is not possible. Thus $\boxed{2b \in \mathbb{Z}}$.

Let $2a = u, 2b = v$. Then from $a^2 - b^2 d \in \mathbb{Z}$ we have $\frac{u^2 - v^2 d}{4} \in \mathbb{Z}$. That is $u^2 - dv^2 \equiv 0 \pmod 4$.

$d \equiv 2$ *or* $3 \pmod 4$ Then $u^2 + 2v^2 \equiv 0 \pmod 4$ or $u^2 + v^2 \equiv 0 \pmod 4$ as $2 \equiv -2 \pmod 4$ and $3 \equiv -1 \pmod 4$. Thus it follows that u, v are even or $a, b \in \mathbb{Z}$. Equivalently $O_{\mathbb{K}} = \mathbb{Z}[\sqrt{d}]$.

$d \equiv 1 \pmod 4$ $4|u^2 - dv^2 \Rightarrow 4|u^2 - v^2 \Rightarrow 4|(u-v)(u+v)$. That is u, v have same parity (both even or both odd). Thus $O_{\mathbb{K}} = \{\frac{u+v\sqrt{d}}{2} | u \equiv v \pmod 2\}$. Notice that $\frac{u+v\sqrt{d}}{2} = \frac{u-v}{2} + \frac{1+\sqrt{d}}{2}v$.

$\square$

* As we know every quadratic field over $\mathbb{Q}$ is of the form $\mathbb{Q}[\sqrt{d}]$ for some square free rational integer d. If $d < 0$ then $\mathbb{Q}[\sqrt{d}]$ is called **imaginary quadratic number field** otherwise it is called **real quadratic number field.**

* Let $\mathbb{K}$ be a number filed, then it is easy to verify $O_{\mathbb{K}} = \mathbb{K} \cap \overline{\mathbb{Z}}$ is a subring of $\mathbb{K}$ called **ring of integers** of $\mathbb{K}$. From Theorem 7.11 we have

	Number field ($\mathbb{K}$)	Ring of Integers ($O_{\mathbb{K}}$)
1	$\mathbb{Q}$	$\mathbb{Z}$
2	$\mathbb{Q}[i] = \mathbb{Q}[\sqrt{-1}]$	$\mathbb{Z}[i]$
3	$\mathbb{Q}[\sqrt{5}]$	$\mathbb{Z}\left[\frac{[1+\sqrt{5}}{2}\right]$
4	$\mathbb{Q}[\sqrt{-5}]$	$\mathbb{Z}[\sqrt{-5}]$
5	$\mathbb{Q}[\sqrt{2}]$	$\mathbb{Z}[\sqrt{2}]$
6	$\mathbb{Q}[\sqrt{-3}]$	$\mathbb{Z}\left[\frac{[1+\sqrt{-3}}{2}\right]$

Theorem 7.12. *Let $d < 0$ be a square free rational integer.*

1. *Then ± 1 are units in $O_{\mathbb{Q}[\sqrt{d}]}$. These are the only units in $O_{\mathbb{Q}[\sqrt{d}]}$ if $d \neq -1, -3$.*

2. *The units in $O_{\mathbb{Q}[\sqrt{-1}]}$ are $\{1, -1, i, -i\}$.*

3. *The units in $O_{\mathbb{Q}[\sqrt{-3}]}$ are $\{1, -1, \omega, -\omega, \omega^2, -\omega^2\}$, where $\omega = \frac{1+\sqrt{-3}}{2}$.*

Theorem 7.13. *There are infinite number of units in any ring of integers of real quadratic filed.*